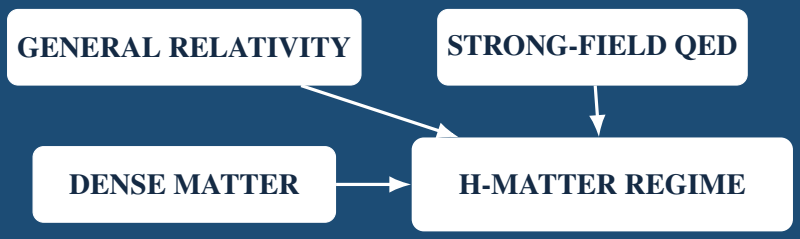
THE HYPER-DENSE MAGNETO-GRAVITATIONAL QUANTUM MATTER PROJECT

HMGQM Theory · Volume 3 Formulation

Volume 3

Strong-field QED + dense matter + classical general relativity
with anisotropic stresses in an axisymmetric compact-object framework



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Research formulation

Scientific Status and Reader Notice

Status of this document

This document is a consolidated theoretical research framework, not an experimentally verified new law of nature. It intentionally separates established physical ingredients from the original H-Matter hypothesis. The classical gravity sector, strong-field QED effective action, Landau quantization, magnetization, electromagnetic stress, and relativistic dense-matter microphysics are established components. The distinctive scientific question is whether their self-consistent combination creates a quantitatively new regime, stable equilibrium branch, or observational signature that cannot be reproduced by existing magnetized compact-star models.

The original personal draft introduced H-Matter as a proposed extreme matter phase generated by the simultaneous action of dense matter, ultra-strong magnetic fields, spacetime curvature, and quantum-vacuum effects. [1]

The present edition preserves that physical intuition while tightening the domain of validity and the mathematical logic. In particular, it does not assume quantum gravity, does not equate $B \gg B_{\mathrm{QED}}$ with one-dimensionality of all baryons, does not use a weak-field F^4 truncation as the defining strong-field QED model, and does not claim that local inequalities alone establish global stellar stability. These are central structural decisions of the current formulation. [2]

Abstract

HMGQM Theory v3.1 develops a self-consistent effective framework for the study of ultra-magnetized, ultra-dense compact matter in classical general relativity. The central hypothesis is that, inside a sufficiently magnetized compact object, Landau quantization of charged particles, quantum-electrodynamic vacuum response, electromagnetic stress, dense-matter microphysics, and spacetime curvature can all become dynamically important at once. In that regime, the local stress tensor is intrinsically anisotropic and the global equilibrium is generically axisymmetric rather than spherical.

The formulation begins from a covariant action containing the Einstein-Hilbert term, the Maxwell term, a renormalized finite-field one-loop Heisenberg-Euler correction, and an explicitly specified dense-matter model. The local thermodynamics is represented by a grand potential $\Omega(\{\mu_i\}, T, B)$, from which magnetization and the principal pressures are derived. Charged species are quantized into Landau levels, while neutral species and anomalous magnetic moments are treated separately. The total stress-energy tensor is defined once, from the action, so Maxwell and QED vacuum contributions are not inserted a second time by hand.

On macroscopic scales the theory is solved through the coupled axisymmetric Einstein-Maxwell-matter equations with a specified magnetic topology and boundary conditions. Stability is treated hierarchically: local thermodynamic consistency, global equilibrium diagnostics, and finally a linear dynamical eigenvalue problem. Observable sectors include mass-radius structure, moments of inertia, tidal deformability, magnetic ellipticity, gravitational-wave mode shifts, and polarization-dependent photon transport. A thermodynamic phase claim is intentionally stronger than the regime label H-Matter: it requires a non-analytic thermodynamic feature or a well-defined order parameter.

The final scientific position is therefore deliberately falsifiable. HMGQM is not claimed to replace general relativity or quantum electrodynamics. It is a proposed synthesis and extreme-regime diagnostic whose scientific value will be determined by derivation, numerical convergence, stability analysis, and observational discrimination.

Keywords

- Heisenberg-Euler effective action
- magnetized dense matter
- Landau quantization and anomalous magnetic moments
- Einstein-Maxwell equilibrium
- vacuum birefringence
- anisotropic relativistic stresses
- magnetar interiors
- compact-object stability
- falsifiable astrophysical observables

Note About the Author

About the author

Anish Kushwaha is an independent student and researcher with interdisciplinary interests spanning theoretical physics, cosmology, computation, cybersecurity, and analytical problem solving. In the original HMGQM draft, the work was explicitly presented as an exploratory framework intended for refinement and critical examination rather than as an experimentally verified result. [3]
The present document follows the same learning-oriented position: the theory is treated as a hypothesis that must survive consistency checks before stronger claims are justified. The authorś interest is centered on regimes where extreme astrophysical conditions force several physical descriptions to operate together, especially general relativity, strong-field QED, and dense-matter theory.

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How to Read This Edition

This document is organized as a research-style monograph rather than a short proposal. Part I reconstructs the intellectual path from the original draft to the present framework. Part II establishes the mathematics. Part III treats the dense-matter and QED microphysics. Part IV constructs global equilibrium. Part V addresses stability. Parts VI and VII develop the electromagnetic and gravitational-wave observable sectors. Part VIII is a numerical implementation program. Part IX defines falsifiability and the boundary between evidence for a model effect and evidence for a genuinely new phase. The appendices collect derivations, dimensionless variables, numerical pseudocode, and reference notes.

Throughout, statements are intentionally classified as one of three types: **established physics**, **model assumptions**, or **HMGQM hypotheses**. A diagram labeled *schematic* is an explanatory representation and not a fitted observational result.

Contents

Abstract ii

Keywords iii

Note About the Author iv

How to Read This Edition v

Notation and Conventions xvii

I From the Original Hypothesis to a Defensible Theory 1

1 Introduction and Physical Motivation 2
1.1 The original conceptual question . . . 2
1.2 Why the problem is physically interesting . . . 2
1.3 The central hypothesis in its modern form . . . 2

2 From v1.0 to v3.1: What Changed and Why 4
2.1 Original architecture . . . 4
2.2 The decisive contradictions . . . 4
2.3 Repair map . . . 4

3 Scientific Positioning and Novelty Boundary 6
3.1 Established ingredients . . . 6
3.2 The potential novelty . . . 6

4 Domain of Validity 7
4.1 Field and density scales . . . 7
4.2 Operational H-regime diagnostics . . . 7
4.3 Curvature diagnostic . . . 7

II Research Atlas: Established Results Surrounding HMGQM 8

5 Why the Research Context Matters 9

6 Magnetic-Field Scale Atlas 10

7 Landau-Level Fan: The Microscopic Origin of Anisotropy 11

8 Landau Occupancy and the H-Regime Diagnostic 12

9 Species-Resolved Sensitivity 13

10 Magnetic Geometry Atlas 14

11 Magnetic Force-Balance Logic 15

12 Current Closure and the Electromagnetic Sector **16**

13 Surface Matching and the Exterior Problem **17**

14 A Research-Informed Polarization Benchmark **18**

15 GW170817: The Correct Observational Interpretation **19**

16 Baseline-vs-HMGQM Comparison **20**

17 Parameter-Space Map for the H-Regime **21**

18 Verification Matrix for a Trustworthy Result **22**

19 Research Claim Ladder **23**

20 Research Evidence Matrix **24**

21 From Local Physics to an Astrophysical Observable **25**

22 The Three-Dimensional HMGQM Research Space **26**

23 Research Synthesis: What the Literature Already Tells Us **27**
23.1 Strong-field QED 27
23.2 Magnetized dense matter 27
23.3 Relativistic magnetic equilibria 27
23.4 Polarization observations 27
23.5 Gravitational-wave discipline 27

III Research-Informed Expansion and Diagram Atlas **29**

24 Scale Separation: Microphysics, Macrophysics, and the H-Regime **30**

25 Explicit Landau Spectrum and Level Occupation **31**

26 Which Species Feel the Field First? **32**

27 Magnetic Stress versus Matter Stiffness **33**

28 Grand Potential to Observable Stress **34**

29 Stress Tensor Geometry in the Local Rest Frame **35**

30 Poloidal Magnetic Topology **36**

31 Toroidal Magnetic Topology **37**

32 Mixed Poloidal-Toroidal Configuration **38**

33 Axisymmetric Star Cross-Section **39**

34 Surface Matching and the Exterior Magnetosphere **40**

35 Finite-Field Birefringence as a High-Value Test Channel **41**

36 Gravitational-Wave Mode Accounting **42**

37 Tidal and Rotational Diagnostics **43**

38 Parameter Degeneracy and Model Comparison **44**

39 Numerical Convergence as a Scientific Requirement **45**

40 Uncertainty Propagation Pipeline **46**

41 Regime Label versus Phase Discovery **47**

42 The Evidence Ladder **48**

43 Falsification Decision Tree **49**

44 Reproducibility and Independent Reimplementation **50**

45 Research Update: What Changed the Standard of Proof **51**

IV Covariant Mathematical Formulation **52**

46 Units, Conventions, and Variational Structure **53**
46.1 Natural-unit convention 53
46.2 The defining action 53

47 Strong-Field QED: Finite-Field Heisenberg-Euler Sector **54**
47.1 Why the weak-field truncation is not the defining model 54
47.2 Renormalized proper-time representation 54
47.3 Asymptotic strong-field behavior 54
47.4 Constitutive tensor 55

48 Einstein Equation and Stress-Energy Accounting **56**
48.1 Local anisotropic decomposition 56
48.2 Conservation law 57

V Microscopic Dense Matter and Magnetized Thermodynamics **58**

49 Particle Content and Chemical Equilibrium **59**
49.1 Minimal reference composition 59
49.2 Why the composition matters 59

50 Landau Quantization **60**
50.1 Charged relativistic fermions 60
50.2 Zero-temperature grand potential 60
50.3 No automatic one-dimensional collapse 60

51 Neutral Species and Anomalous Magnetic Moments **62**

52 Thermodynamic Potentials, Magnetization, and Principal Stresses 63
52.1 Grand potential . . . 63
52.2 Energy density . . . 63
52.3 Principal pressures . . . 63
52.4 A warning about signs and conventions . . . 63

53 H-Matter as a Regime and as a Possible Phase 65
53.1 Regime definition . . . 65
53.2 Phase criterion . . . 65

VI Axisymmetric General-Relativistic Equilibrium 66

54 Why Spherical TOV Is Not the Final Equation 67

55 Metric Ansatz and Magnetic Geometry 68
55.1 Stationary axisymmetry . . . 68
55.2 Four-potential . . . 68

56 Einstein-Maxwell-Matter Boundary-Value Problem 69
56.1 Boundary conditions . . . 69

57 Horizon-Free Domain and Compactness 70

VII Stability, Causality, and Perturbation Theory 71

58 Local Thermodynamic Consistency 72

59 Global Equilibrium Diagnostics 73

60 Dynamical Stability 74

61 Mode Classification 75

VIII Observable Consequences 76

62 Vacuum Birefringence and Polarized Photon Propagation 77
62.1 Established QED effect . . . 77
62.2 Finite-field response . . . 77

63 Polarized Ray Tracing in Curved Spacetime 78

64 Gravitational-Wave Observables 79
64.1 GW170817 and the post-merger caution . . . 79
64.2 Potential model outputs . . . 79

65 Mass-Radius, Tidal, Rotational, and Electromagnetic Diagnostics 80

IX Numerical Formulation and Research Workflow 81

66 Dimensionless Formulation 82

67 Numerical Architecture **83**

68 Convergence and Error Control **84**

69 Parameter-Space Studies **85**

70 Benchmark Problems **86**

71 Reproducibility Protocol **87**

72 Microphysics Module Architecture **88**
72.1 Reference EOS versus universal HMGQM physics 88

73 Species-Resolved Landau Occupancy **89**

74 Magnetic Stress Budget **90**

75 Field-Topology Dependence **91**

76 Surface Matching and Exterior Propagation **92**

77 Thermodynamic Branch Analysis **93**

78 Uncertainty Budget **94**

79 Verification Pyramid **95**

80 Multi-Messenger Consistency Triangle **96**

81 Model-Selection Logic **97**

82 A Reproducibility Manifest **98**

83 Worked Symbolic Consistency Check **99**

X Falsifiability and Scientific Tests **100**

84 Observational Discipline **101**
84.1 Baseline comparison principle 101

85 Failure Conditions **102**

86 Potentially Positive Evidence **103**

87 Research Program **104**

XI Conclusions and Final Scientific Position **105**

88 Integrated Theory Statement **106**

89 Limiting Cases **107**
89.1 Zero-field limit 107
89.2 Weak-field QED limit 107
89.3 Many-Landau-level limit 107

89.4 Weak magnetic-stress limit . . . 107
89.5 Newtonian limit . . . 107

90 Conclusion **108**

91 What This Theory Does Not Claim **109**

Closing Statement **110**

Appendices **112**

A Landau-Level Derivation Details **112**
A.1 Lowest-Landau-level criterion . . . 112

B Finite-Field QED Checks **113**

C Stress Tensor Identities **114**

D Dimensionless Research Variables **115**

E Suggested Numerical Pseudocode **116**

F Benchmark Table **117**

G Observational Context Notes **118**
G.1 Magnetar polarization . . . 118
G.2 GW170817 . . . 118

H Reference Notes and Research Basis **119**

I Expanded Comparison with the Original 23-Page Draft **120**

Author Contact **121**

References **122**

List of Figures

1.1 Conceptual structure of HMGQM: three established physical sectors enter one coupled calculation. The label H-Matter is a regime hypothesis until a non-continuous phase criterion is demonstrated. 2

2.1 Evolution of the framework. v3.1 expands the corrected v2.0 structure rather than returning to the stronger unsupported claims of the original draft. 4

4.1 Schematic hierarchy of regimes. HMGQM is not defined by a single numerical threshold; it is an operational intersection of high field, high density, important Landau structure, and non-perturbative magnetic/QED stress. 7

5.1 Research-context architecture. The scientific novelty sought by HMGQM lies in the coupled solution and in any genuinely new consequence, not in claiming ownership of the underlying GR, QED, or dense-matter ingredients. 9

6.1 Field-scale atlas synthesized from the cited magnetar and compact-star literature. The shaded segments indicate commonly discussed scales, not a universal stellar field profile or an observational determination of interior B. 10

7.1 Landau-level fan for a free relativistic charged fermion at zero longitudinal momentum. The figure is derived directly from the level spectrum used by HMGQM and illustrates why a three-dimensional momentum integral is incomplete when only a few Landau levels are occupied. 11

8.1 Representative Landau-occupancy benchmark for $\mu^*/m^* = 10$ and unit charge. The integer staircase is illustrative; the actual HMGQM calculation solves charge neutrality, beta equilibrium and the interacting EOS simultaneously. 12

9.1 Species-resolved matter response. A single field amplitude does not define a universal microscopic response because charge, mass, chemical potential and magnetic moment differ among species. 13

10.1 Original schematic atlas for three idealized magnetic topologies. The real HMGQM calculation uses the chosen vector potential and current function; these drawings are conceptual topology guides. 14

11.1 Force-balance logic. Rotation is optional; in a nonrotating configuration its contribution vanishes. The final equilibrium statement is covariant conservation, not a hand-assembled sum of forces. 15

12.1 Electromagnetic closure cycle. The constitutive tensor $H^{\mu\nu}$ packages the finite-field QED response; the current and boundary conditions complete the global magnetic problem. . . 16

13.1 Interior-to-observer chain. The final observable is an integrated property of the interior, surface, magnetosphere and propagation geometry. 17

14.1 Original summary graphic of the polarization anchors reported in the 2026 Nature observation. It is not a reproduction of the paper's figure and does not invent a phase curve; it only shows the reported magnitude ranges and their observational role in the HMGQM comparison program. 18

15.1 Corrected observational discipline for GW170817. The event is a constraint source, not a direct datum for the mode-frequency relation plotted in the original HMGQM draft. . . . 19

16.1 Baseline comparison principle. A difference from a baseline is evidence for a model effect; it becomes evidence for a new phase only if the effect survives microphysical alternatives and phase diagnostics. 20

17.1 Parameter-space survey map. The boundaries are deliberately illustrative: the actual H-regime is defined by the joint diagnostics χ_B, $\nu_{\max,i}$, β_B, density, and the relative electromagnetic/QED contribution to stress. 21

18.1 Verification gate. A beautiful diagram or a converged number is not sufficient unless the equations, units, conservation laws and numerical errors also close consistently. 22

19.1 Claim ladder. H-Matter begins as a regime label; a stronger phase claim is the final step and requires thermodynamic or order-parameter evidence. 23

20.1 Research evidence matrix. The model is strongest when it explicitly separates what has already been demonstrated in the literature from what remains a falsifiable HMGQM question. 24

21.1 End-to-end physical hierarchy. The same formal structure appears repeatedly in the theory: local microphysics feeds global geometry, which feeds the observable sector. 25

22.1 Research-space schematic. HMGQM is a multidimensional boundary-value problem; field amplitude alone is not a sufficient coordinate for its physical classification. 26

24.1 Scale hierarchy for the HMGQM program. Each layer has its own equations and uncertainties; the proposed theory becomes predictive only when information is passed consistently between layers. 30

25.1 Landau-level fan schematic. The Fermi surface intersects only those levels with real longitudinal Fermi momentum. Stronger B increases the spacing between levels, but the lowest-level regime must still be checked species by species. 31

26.1 Species sensitivity hierarchy. The figure is deliberately schematic: exact thresholds depend on chemical potentials, interactions, anomalous magnetic moments, and composition. Neutrons are neutral and are not Landau quantized, although their magnetic moments can still couple to the field. 32

27.1 Stress-budget schematic. A crossover between matter and magnetic stresses is not itself a phase transition; it identifies a regime in which magnetic geometry can no longer be treated as a perturbation of an isotropic star. 33

28.1 Thermodynamic Legendre map. The grand potential is not an ornament: it is the bridge from microscopic composition to a consistent stress-energy closure. 34

29.1 Local stress geometry. When $P_{\parallel} \neq P_{\perp}$, the material sector cannot be represented by a single scalar pressure. 35

30.1 Poloidal field topology. The field lines thread the object and extend into the exterior. The exact topology is determined by the current distribution and boundary conditions, not by the H-Matter label alone. 36

31.1 Toroidal-field schematic. Toroidal configurations are a distinct magnetic geometry and can redistribute magnetic energy and deformation differently from poloidal fields. 37

32.1 Mixed topology. HMGQM should ultimately be tested across a family of magnetic geometries because topology is part of the physical state, not just a visualization choice. . 38

33.1 Conceptual meridional section of an axisymmetric compact object. Isodensity contours, field lines, and the symmetry axis form a coupled geometry. 39

34.1 Interior-to-exterior transport chain. Two photon polarization modes can follow different effective propagation laws in strong magnetic fields. A HMGQM observable must include the exterior transport and surface emission model consistently. [13, 12] 40

35.1 Schematic polarization transport hierarchy. The near-surface region can be the most sensitive to finite-field response, while mode decoupling can occur farther out where the local field has weakened substantially. The exact curves are model dependent. [12] . . . 41

36.1 Mode-spectrum schematic. In an axisymmetric magnetized configuration, degeneracies and mode frequencies can be shifted or split. The numerical shift is a prediction only after the coupled perturbation equations are solved. 42

37.1 Multi-observable diagnostic strategy. HMGQM should be evaluated as a vector of residuals rather than a single curve. 43

38.1 Parameter-degeneracy schematic. An apparent HMGQM residual should not be interpreted as evidence for a new phase when it can be absorbed by plausible EOS or geometry uncertainty. 44

39.1 Convergence schematic. A visually smooth solution is not enough; residuals of the field equations and conservation laws must also decrease under refinement. 45

40.1 Uncertainty propagation pipeline. The HMGQM hypothesis is tested after nuisance uncertainties are propagated, not before. 46

41.1 Scientific decision boundary between an extreme regime and a newly established phase. This distinction is one of the most important safeguards in the revised theory. 47

42.1 Evidence ladder. HMGQM should be considered scientifically stronger as it climbs from formal consistency to independently reproducible observation. 48

43.1 Falsification decision tree. A candidate must pass mathematical consistency, known-limit recovery, numerical convergence, and dynamical-stability checks before any distinctive observational residual can support a stronger claim. 49

44.1 Reproducibility manifest. The feedback arrow represents versioned inputs and preserved numerical diagnostics, enabling independent reimplementation. 50

45.1 Research synthesis map. The literature supplies the ingredients and constraints; HMGQM is a hypothesis about their coupled extreme-regime consequences. 51

46.1 Action architecture. The diagram is the central anti-double-counting rule of the theory. . 53

47.1 Finite-field versus truncated strong-field response. The plot is schematic and is not a numerical fit. . . . 54

48.1 Local principal-stress geometry. . . . 56

49.1 Self-consistent composition loop. The dashed arrow represents feedback through equilibrium and the EOS. . . . 59

50.1 Landau-level ladder. The schematic emphasizes that strong fields discretize transverse motion while leaving a continuous longitudinal momentum p_z. . . . 61

52.1 Thermodynamic map: one grand potential generates energy, magnetization, and principal stresses. . . . 64

53.1 Schematic phase criterion. A regime can be smoothly connected to ordinary matter; a phase claim requires additional thermodynamic structure. . . . 65

55.1 Magnetic geometry is part of the physics. The same local field amplitude can lead to different global deformations for different topologies. . . . 68

56.1 Self-consistent equilibrium loop. Geometry, field, microphysics, and boundaries are solved together. . . . 69

57.1 Compactness and horizon-free reasoning. . . . 70

59.1 Stability hierarchy. Each stage answers a different question. . . . 73

61.1 Schematic mode spectrum. The red point represents a mode with positive growth rate under the chosen convention. . . . 75

62.1 Polarization-dependent photon propagation in a magnetized vacuum. The curves are schematic, not numerical predictions. . . . 77

63.1 Research pipeline for polarization observables. . . . 78

64.1 Illustrative mode-spectrum shift. Not a fit to LIGO data; the point is that a calculation must predict the shift from the perturbation operator. . . . 79

65.1 Mass-radius comparison as a diagnostic family rather than an assumed HMGQM signature. 80

67.1 Numerical architecture from microphysics to observables. . . . 83

68.1 Convergence ladder. A stable-looking picture without convergence is not a validated result. 84

69.1 H-regime parameter-space map. No fixed universal boundary is assumed. . . . 85

72.1 Modular microphysics. Each block can be changed or benchmarked without rewriting the global GR structure. . . . 88

73.1 Species-resolved Landau occupancy. . . . 89

74.1 Local magnetic stress budget as a diagnostic, not a fixed partition. . . . 90

75.1 Topology family. . . . 91

76.1 Interior-to-exterior matching. . . . 92

77.1 Schematic equilibrium-branch analysis. 93

78.1 Suggested uncertainty budget. 94

79.1 Verification pyramid for a reproducible implementation. 95

80.1 Multi-messenger consistency triangle. Correlated predictions are stronger than isolated phenomenology. 96

81.1 Model-selection logic. Each step narrows the space of defensible claims. 97

83.1 Symbolic consistency-check concept for thermodynamic derivatives. 99

84.1 Baseline-comparison principle for falsifiable predictions. 101

86.1 Decision tree for turning an H-regime calculation into defensible scientific evidence. . . 103

87.1 End-to-end research pipeline. 104

89.1 Required limiting-case consistency. 107

Notation and Conventions

Symbol	Meaning
$g_{\mu\nu}$	spacetime metric; signature $(-,+,+,+)$
R	Ricci scalar
$F_{\mu\nu}$	electromagnetic field tensor
$\tilde{F}_{\mu\nu}$	dual electromagnetic tensor
u^{μ}	fluid four-velocity
b^{μ}	unit magnetic-field direction in the comoving frame
B	local magnetic-field amplitude in the chosen frame
μ_i	chemical potential of species i
m_i^*	effective mass supplied by the chosen interacting EOS
Ω	renormalized grand-potential density
M	magnetization, $M = -\partial\Omega/\partial B$
$P_{\parallel}$	total principal stress parallel to B
$P_{\perp}$	total principal stress transverse to B
B_{QED}	electron QED critical field, approximately 4.414×10^9 T
n_B	baryon number density
ν	Landau-level index
$\nu_{\max}$	largest occupied Landau level
χ_B	dimensionless magnetic-field strength B/B_{QED}
β_B	magnetic-stress to matter-pressure ratio
χ_R	local curvature diagnostic relative to the electron Compton scale
Λ	dimensionless tidal deformability
ϵ_B	magnetic ellipticity/deformation parameter

Part I

From the Original Hypothesis to a Defensible Theory

1

Introduction and Physical Motivation

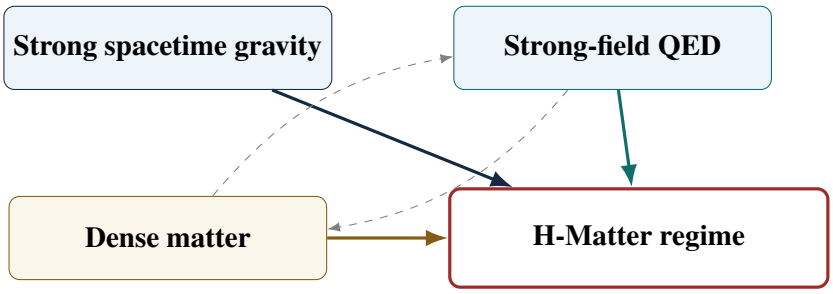
1.1 The original conceptual question

The original HMGQM draft posed a simple but ambitious question: what happens when the density of matter is so high that relativistic degeneracy pressure matters, the magnetic field is far above the electron QED critical field, and the spacetime curvature is strong enough that a compact object cannot be treated as a weakly perturbed Newtonian system? The draft described this environment as the birthplace of a proposed anisotropic matter regime, H-Matter. The defining intuition was that no single scalar equation of state can capture the complete local stress structure when charged-particle motion, vacuum polarization, and electromagnetic stress become direction-dependent. [1]

1.2 Why the problem is physically interesting

Magnetars provide a natural astrophysical laboratory for supercritical magnetic fields. Surface fields around 10^{14}-10^{15} G are inferred for magnetars, while interior fields may be much larger and are often modeled up to the 10^{17}-10^{18} G range in theoretical studies. Strong fields modify composition, quantize charged-particle motion, introduce magnetization, and deform the global stellar geometry. [9, 10, 7, 8]

The important logical point is that these effects are not independent knobs. A larger field changes the Landau-level population; the changed composition modifies the matter pressure; magnetization changes the principal stresses; those stresses deform the star; the deformed geometry changes the field configuration and photon propagation; and the entire system must satisfy a single covariant conservation law.



The present theory does not assume a new force. It asks whether the coupled feedback among these established sectors creates a distinct, dynamically important regime.

Figure 1.1: Conceptual structure of HMGQM: three established physical sectors enter one coupled calculation. The label H-Matter is a regime hypothesis until a non-continuous phase criterion is demonstrated.

1.3 The central hypothesis in its modern form

The strongest defensible statement of the theory is:

Central HMGQM hypothesis

In a sufficiently dense, strongly magnetized compact object, Landau quantization of charged particles, QED vacuum response, electromagnetic stress, and relativistic gravity can become simultaneously non-perturbative relative to the local matter balance. The resulting stress-energy is anisotropic and the equilibrium must be solved self-consistently in axisymmetry. The regime is denoted H-Matter.

This statement is deliberately narrower than the original language of a "quantum-gravitational phase of spacetime." The present formulation keeps the gravitational sector classical and treats a Planck-scale quantum-gravity completion as outside the model. [2]

2

From v1.0 to v3.1: What Changed and Why

2.1 Original architecture

The original draft proposed a joint limit involving ultra-high curvature, ultra-strong magnetic fields, dense fermionic matter, vacuum stress, and anisotropic pressure. It introduced an action, a stress tensor, an anisotropic EOS, a modified TOV equation, an axisymmetric metric, stability inequalities, and several observational ideas. [1]

2.2 The decisive contradictions

The largest conceptual contradiction was the simultaneous use of Planck-scale or non-perturbative quantum-gravity language and an explicit assumption that quantum gravity was not included and that the model remained below the Planck regime. Another problem was the use of a finite F^4 Heisenberg-Euler truncation in a regime explicitly defined as $B \gg B_{\mathrm{QED}}$. A third was the absence of an actual Landau-level sum in the purportedly Landau-quantized EOS. [1, 2]

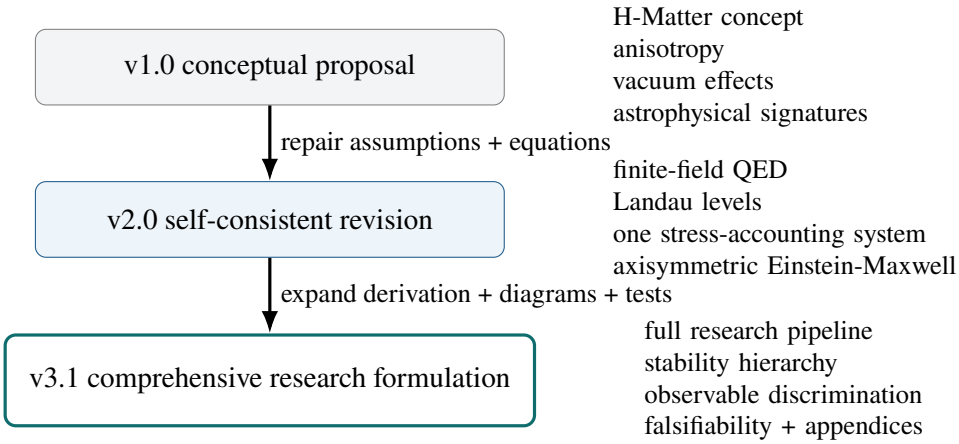


Figure 2.1: Evolution of the framework. v3.1 expands the corrected v2.0 structure rather than returning to the stronger unsupported claims of the original draft.

2.3 Repair map

Original issue	Status	Current rule
Planck-scale quantum geometry	Removed	Classical GR is retained; quantum gravity is outside scope.
Weak-field QED polynomial at $B \gg B_{\mathrm{QED}}$	Replaced	Use finite-field one-loop Heisenberg-Euler response.
Three-dimensional Fermi integral	Replaced	Charged species use explicit Landau sums and cutoffs.

All baryons forced into 1D filaments	Rejected	Landau quantization is species-dependent; neutral particles require separate treatment.
Spherical anisotropic TOV	Restricted	General equilibrium is axisymmetric Einstein-Maxwell-matter.
Repeated magnetic/vacuum stresses	Fixed	One action defines one total $T_{\mu\nu}$.
Unspecified $m^*(\rho, B)$	Fixed	Effective masses belong to the chosen microphysical EOS.
"Stable iff" local inequalities	Rejected	Stability requires perturbation/eigenmode analysis.
GW170817/GW190814 mode-frequency points	Removed	Use only actually measured quantities; GW170817 post-merger was not detected. [14]
Anisotropy as proof of a new phase	Rejected	A phase claim requires non-analytic thermodynamics or an order parameter.

3

Scientific Positioning and Novelty Boundary

3.1 Established ingredients

The theory is built from ingredients already present in the literature: the one-loop Heisenberg-Euler effective action, relativistic magnetized dense matter with Landau quantization and magnetization, Maxwell stress, and fully relativistic axisymmetric Einstein-Maxwell equilibrium. [4, 5, 6, 7, 8, 10, 11]

3.2 The potential novelty

The novel content cannot simply be "strong magnetic fields exist" or "pressure becomes anisotropic." Those are known consequences in appropriate regimes. The potential novelty is the *specific coupled diagnostic*: identify a regime where the magnetic/QED terms become order unity in the local stress balance while Landau-level structure affects composition and where these effects feed back into a fully relativistic axisymmetric equilibrium. A new phase would be an additional result, not a starting assumption.

Novelty discipline

A quantitative difference from a baseline magnetized neutron-star model is evidence for a model effect. It is not automatically evidence for a new state of matter. The phase label should be upgraded only if the model produces a thermodynamic non-analyticity, symmetry-breaking order parameter, or other non-continuous structural diagnostic.

4

Domain of Validity

4.1 Field and density scales

The electron QED critical field is

$$B_{\mathrm{QED}} = \frac{m_e^2 c^3}{e\hbar} \simeq 4.414 \times 10^9\ \mathrm{T} \simeq 4.414 \times 10^{13}\ \mathrm{G}. \tag{4.1}$$

The nuclear saturation density is used only as a reference scale, $\rho_{\mathrm{sat}} \simeq 2.8 \times 10^{17}\ \mathrm{kg\,m^{-3}}$. [9]

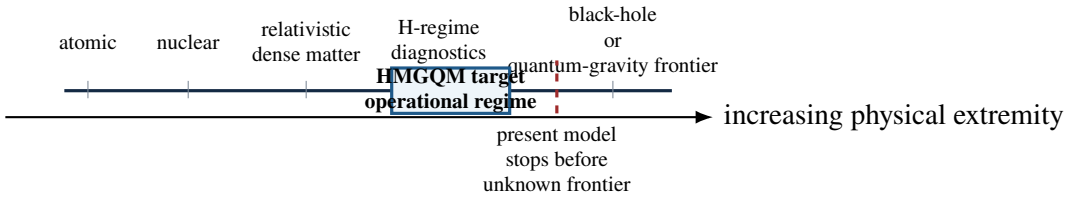


Figure 4.1: Schematic hierarchy of regimes. HMGQM is not defined by a single numerical threshold; it is an operational intersection of high field, high density, important Landau structure, and non-perturbative magnetic/QED stress.

4.2 Operational H-regime diagnostics

Define

$$\chi_B = \frac{B}{B_{\mathrm{QED}}}, \qquad \eta_i = \frac{2|q_i|B}{(\mu_i^*)^2 - (m_i^*)^2}, \qquad \beta_B = \frac{B^2/2}{P_m}, \tag{4.2}$$

where natural units are used and P_m is the matter pressure before explicit Maxwell stress is added. The highest occupied Landau level is

$$\nu_{\mathrm{max},i} = \left\lfloor \frac{(\mu_i^*)^2 - (m_i^*)^2}{2|q_i|B} \right\rfloor. \tag{4.3}$$

A local region is tagged "H-regime" when

$$\begin{aligned} &\chi_B \gg 1, \qquad n_B \gtrsim n_{\mathrm{sat}}, \qquad \beta_B \ll 1, \\ &\nu_{\mathrm{max},i} = O(1) \quad \text{for at least one dynamically important charged species.} \end{aligned} \tag{4.4}$$

This is a regime definition, not a phase proof. [2]

4.3 Curvature diagnostic

Rather than asserting Planck curvature, define the dimensionless local diagnostic

$$\chi_R = \frac{\lambda_e^2}{L_{\mathrm{curv}}^2}, \qquad \lambda_e = \frac{\hbar}{m_e c}, \tag{4.5}$$

where L_{curv} is a curvature radius estimated from an invariant such as $|R|$, $R_{\mu\nu}R^{\mu\nu}$, or the Kretschmann scalar. If $\chi_R \ll 1$, curvature corrections to local electron QED are subleading and the minimal model is local-flat-space QED plus curved-spacetime transport. [2]

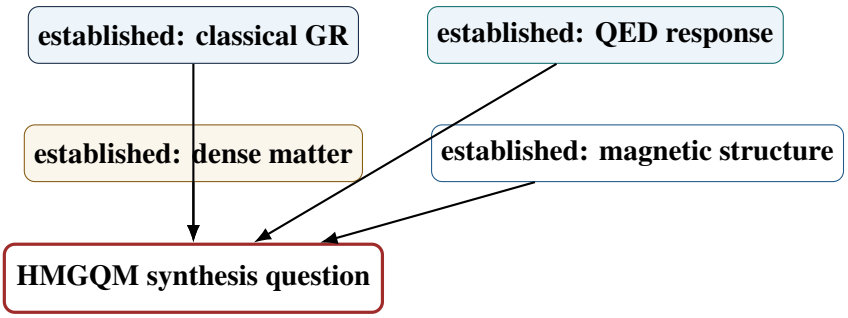
Part II

Research Atlas: Established Results Surrounding HMGQM

5

Why the Research Context Matters

The HMGQM hypothesis sits at the intersection of several research programs that already have independent empirical or theoretical support. A rigorous formulation should therefore be built by identifying what is established, what is model-dependent, and what is specifically hypothesized by HMGQM. Recent literature continues to sharpen this boundary. Strong-field magnetized dense-matter studies investigate Landau quantization, anomalous magnetic moments, composition changes, and the impact of fields on compact-star structure; fully relativistic Einstein-Maxwell calculations show that sufficiently large magnetic stresses deform stars and require axisymmetric self-consistent solutions; and recent X-ray polarization observations of the magnetar 1E 1547.0-5408 provide an important observational context for QED vacuum birefringence. [10, 11, 7, 13]



Does the coupled system create an extreme regime, a stable branch, or a measurable signature that cannot be reproduced by the baseline magnetized-star calculation?

Figure 5.1: Research-context architecture. The scientific novelty sought by HMGQM lies in the coupled solution and in any genuinely new consequence, not in claiming ownership of the underlying GR, QED, or dense-matter ingredients.

6

Magnetic-Field Scale Atlas

The electron QED critical field is $B_{\mathrm{QED}} \simeq 4.414 \times 10^{13}$ G. Magnetar surface fields exceed 10^{14} G, while theoretical interior-field studies often explore 10^{17}-10^{18} G. These are not interchangeable statements: an observed surface field is a measurement, while an interior field is generally model-dependent. [13, 10, 7]

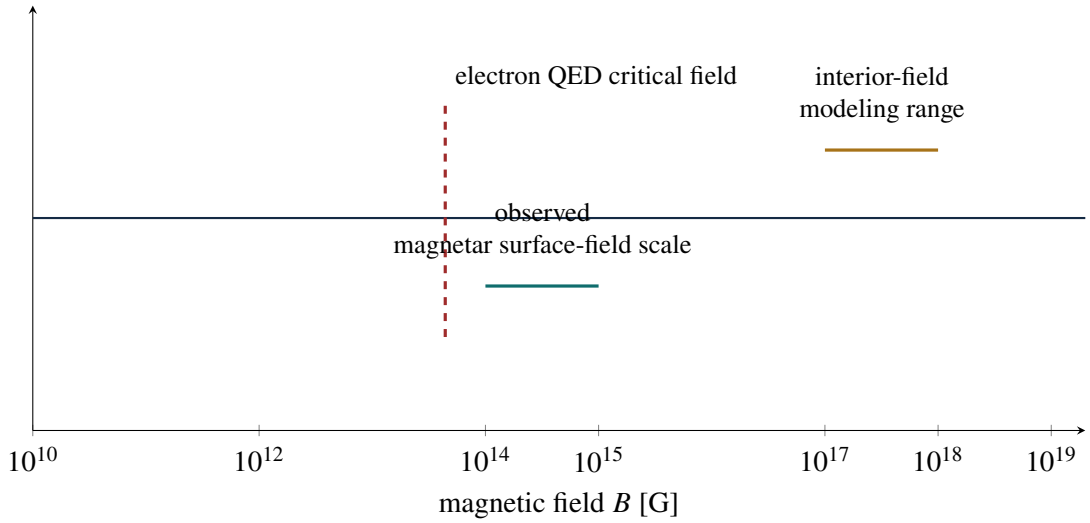


Figure 6.1: Field-scale atlas synthesized from the cited magnetar and compact-star literature. The shaded segments indicate commonly discussed scales, not a universal stellar field profile or an observational determination of interior B.

7

Landau-Level Fan: The Microscopic Origin of Anisotropy

For a charged relativistic fermion in a locally uniform magnetic field, the minimal Dirac spectrum is

$$E_\nu^2 = p_z^2 + (m_i^*)^2 + 2\nu|q_i|B, \qquad \nu = 0, 1, 2, \ldots . \tag{7.1}$$

The transverse motion is quantized while the momentum parallel to the field remains continuous. The following diagram uses the electron mass as a reference and sets $p_z = 0$ solely to visualize the level fan. It is not a stellar spectrum.

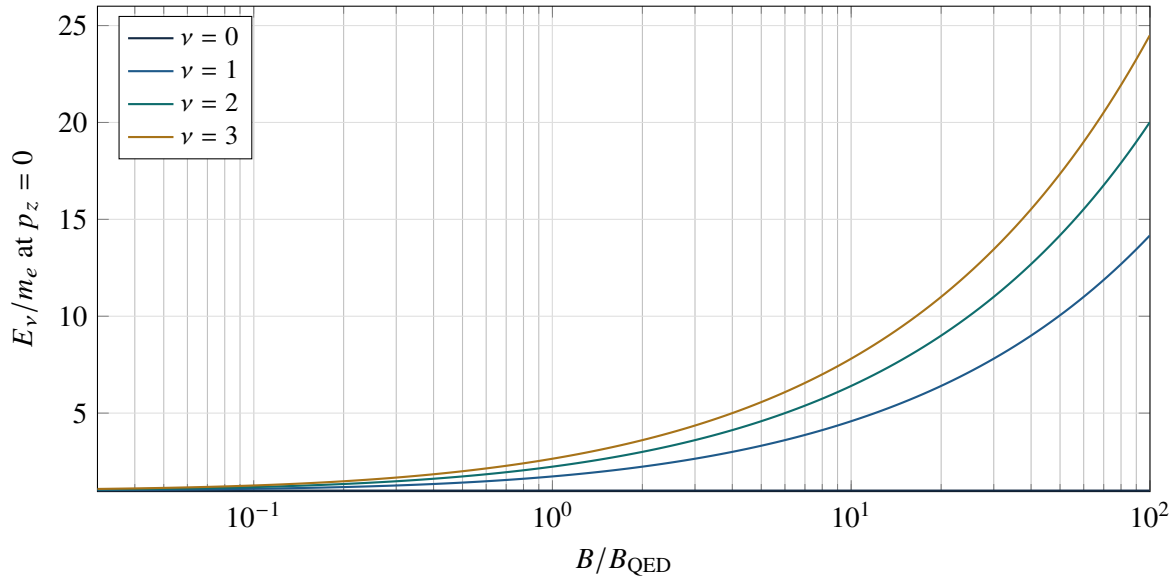


Figure 7.1: Landau-level fan for a free relativistic charged fermion at zero longitudinal momentum. The figure is derived directly from the level spectrum used by HMGQM and illustrates why a three-dimensional momentum integral is incomplete when only a few Landau levels are occupied.

8

Landau Occupancy and the H-Regime Diagnostic

At zero temperature,

$$\nu_{\max,i} = \left\lfloor \frac{(\mu_i^*)^2 - (m_i^*)^2}{2|q_i|B} \right\rfloor . \tag{8.1}$$

As B increases at fixed chemical potential, the number of occupied levels falls in a staircase. Because μ_i^* and m_i^* are themselves determined by the selected interacting EOS, the staircase below is an analytic kinematic benchmark rather than a claim about the composition of any specific star. [16]

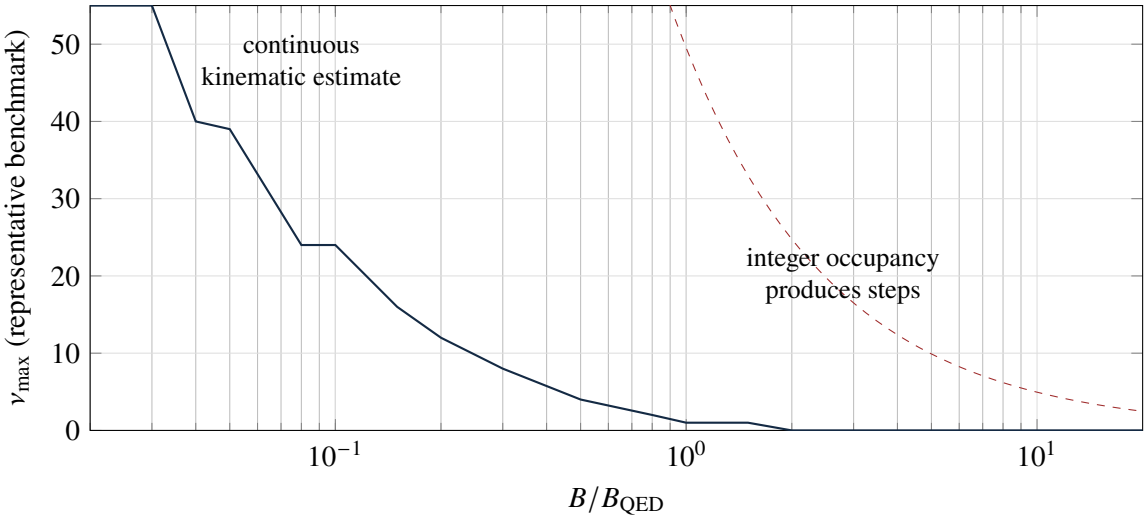
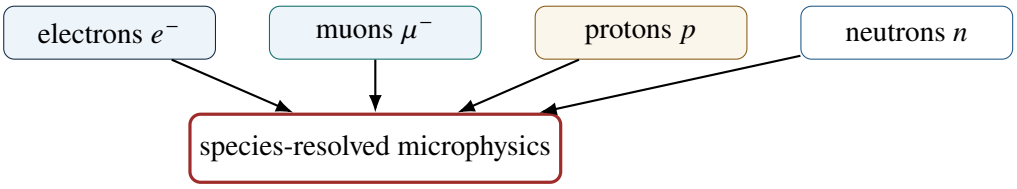


Figure 8.1: Representative Landau-occupancy benchmark for $\mu^*/m^* = 10$ and unit charge. The integer staircase is illustrative; the actual HMGQM calculation solves charge neutrality, beta equilibrium and the interacting EOS simultaneously.

9

Species-Resolved Sensitivity

The field does not act identically on every constituent. Charged leptons and charged baryons are Landau-quantized; neutral baryons are not Landau-quantized in the minimal coupling, although anomalous magnetic moments can still produce spin splitting. Recent hyperonic calculations emphasize the competition between Landau quantization and AMM effects, making a species-by-species treatment essential. [11, 16]



Landau levels + AMM + interactions + chemical equilibrium $\rightarrow$ composition and $\Omega(\{\mu_i\}, T, B)$

Figure 9.1: Species-resolved matter response. A single field amplitude does not define a universal microscopic response because charge, mass, chemical potential and magnetic moment differ among species.

10

Magnetic Geometry Atlas

A local field amplitude is not enough to determine the global solution. Fully relativistic calculations require a magnetic topology and current prescription; different poloidal, toroidal, and mixed configurations can generate different deformations and stress distributions. [7, 17]

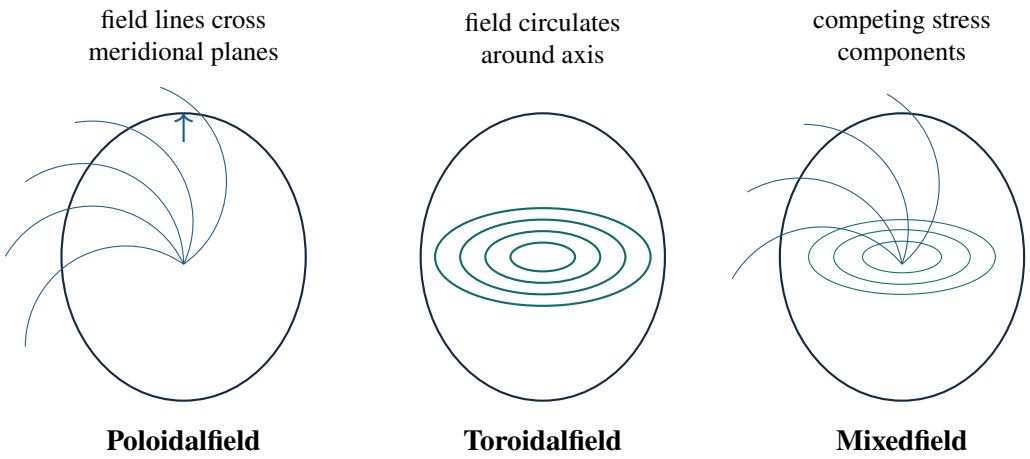


Figure 10.1: Original schematic atlas for three idealized magnetic topologies. The real HMGQM calculation uses the chosen vector potential and current function; these drawings are conceptual topology guides.

11

Magnetic Force-Balance Logic

The coupled equilibrium is easiest to understand as a competition among gravity, matter pressure gradients, electromagnetic stress, magnetization, and rotation when present. Because the total stress tensor is defined variationally, individual force labels are bookkeeping components rather than extra terms that may be added twice.

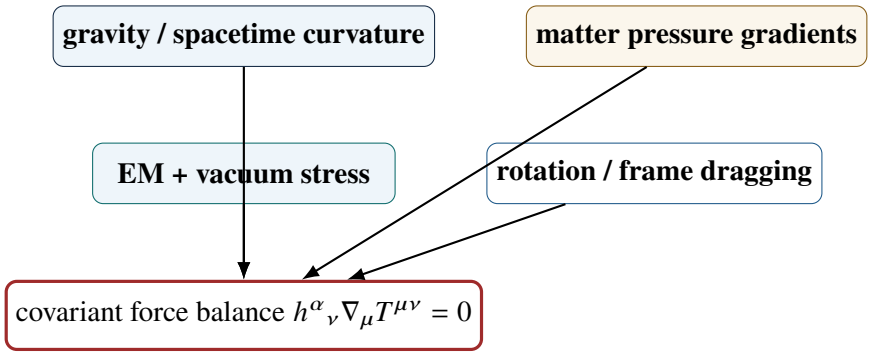


Figure 11.1: Force-balance logic. Rotation is optional; in a nonrotating configuration its contribution vanishes. The final equilibrium statement is covariant conservation, not a hand-assembled sum of forces.

12

Current Closure and the Electromagnetic Sector

The Maxwell system requires not only a field amplitude but a source prescription. In stationary axisymmetry the four-current, the vector potential, the constitutive response, and the boundary data form one closure problem.

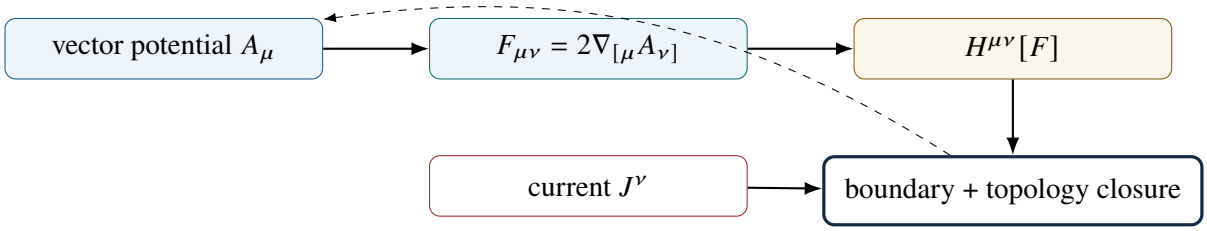


Figure 12.1: Electromagnetic closure cycle. The constitutive tensor $H^{\mu\nu}$ packages the finite-field QED response; the current and boundary conditions complete the global magnetic problem.

13

Surface Matching and the Exterior Problem

An interior H-regime calculation does not end at the stellar surface. Metric matching, electromagnetic matching and radiative transfer connect the interior to an exterior spacetime and magnetosphere. For polarization calculations, a large portion of the observable phase can accumulate outside the dense core. The 2026 magnetar-polarization work makes this especially important because the reported signal is interpreted through magnetospheric propagation rather than through a purely local surface effect. [13, 15]

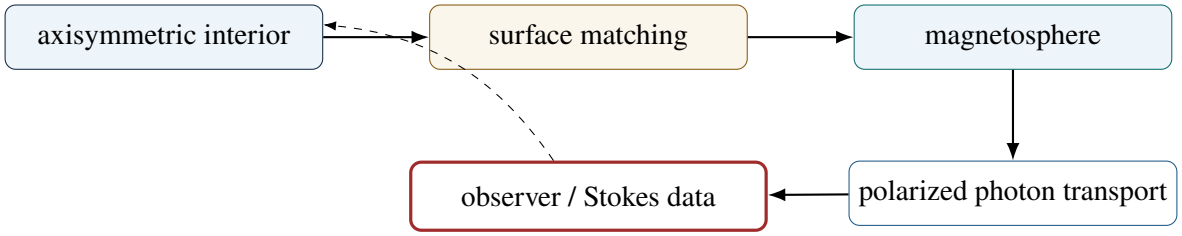


Figure 13.1: Interior-to-observer chain. The final observable is an integrated property of the interior, surface, magnetosphere and propagation geometry.

14

A Research-Informed Polarization Benchmark

The 2026 Nature study of 1E 1547.0-5408 reported high X-ray polarization, including a phase-averaged polarization degree of about 65% at 2 keV and values approaching 80% in selected rotational phases in the 2-3 keV range, while remaining above roughly 40% throughout the radio-beam crossing. The study interpreted the energy and phase dependence as consistent with vacuum-birefringence-governed propagation. These measurements do not establish HMGQM; they establish the observational regime in which a stronger HMGQM calculation must be compared against a QED baseline. [13]

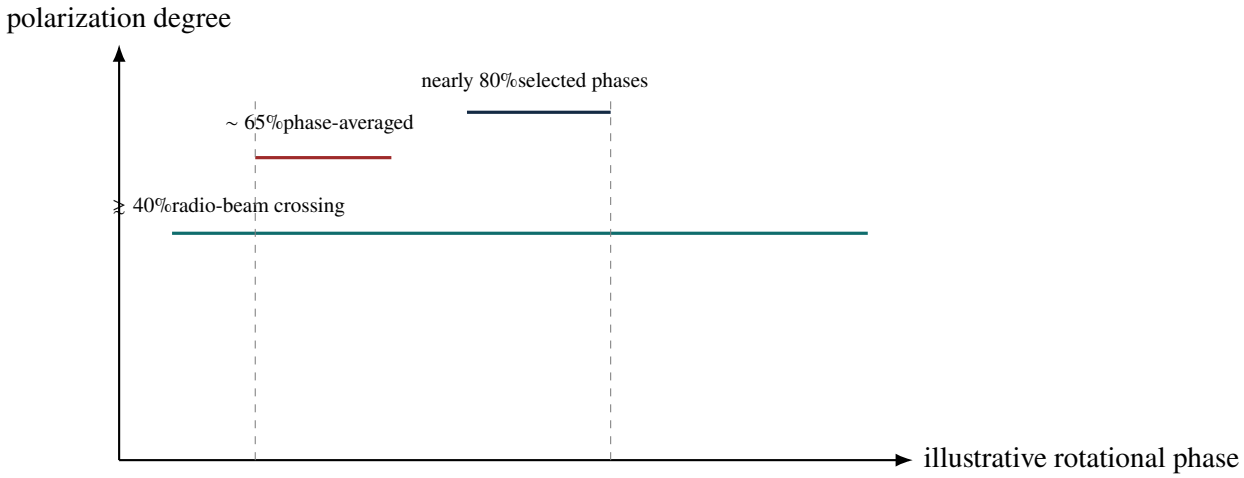


Figure 14.1: Original summary graphic of the polarization anchors reported in the 2026 Nature observation. It is not a reproduction of the paper's figure and does not invent a phase curve; it only shows the reported magnitude ranges and their observational role in the HMGQM comparison program.

15

GW170817: The Correct Observational Interpretation

GW170817 remains relevant to compact-object physics, but the original HMGQM plot must not treat it as a measured post-merger frequency point. LIGO reports that searches for post-merger gravitational waves did not find a statistically significant signal, while the expected short-lived post-merger emission would lie mainly in a high-frequency regime where detector sensitivity was limited. [14, 18]

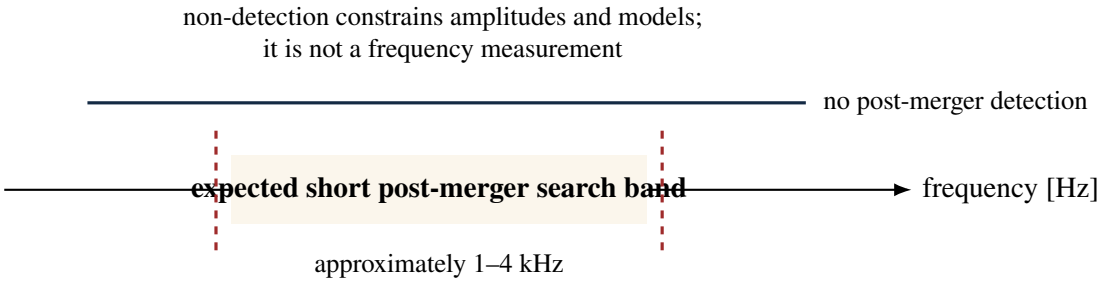
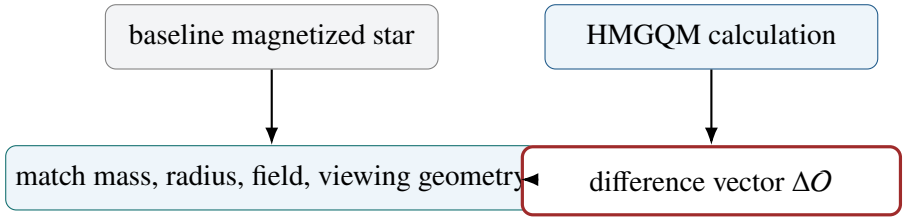


Figure 15.1: Corrected observational discipline for GW170817. The event is a constraint source, not a direct datum for the mode-frequency relation plotted in the original HMGQM draft.

16

Baseline-vs-HMGQM Comparison

A meaningful test must compare HMGQM to a baseline calculation with the same observationally relevant surface quantities and as much of the same microphysics as possible. The goal is to isolate effects caused by the coupled treatment of field geometry, finite-field QED, anisotropic stress and axisymmetric gravity.



Examples: ΔR, $\Delta\Lambda$, $\Delta\omega_{nlm}$, $\Delta\epsilon_B$, and polarization differences after full transfer.

Figure 16.1: Baseline comparison principle. A difference from a baseline is evidence for a model effect; it becomes evidence for a new phase only if the effect survives microphysical alternatives and phase diagnostics.

17

Parameter-Space Map for the H-Regime

The H-regime is intentionally operational rather than defined by a single universal field or density threshold. A useful computational survey therefore spans several dimensionless diagnostics simultaneously.

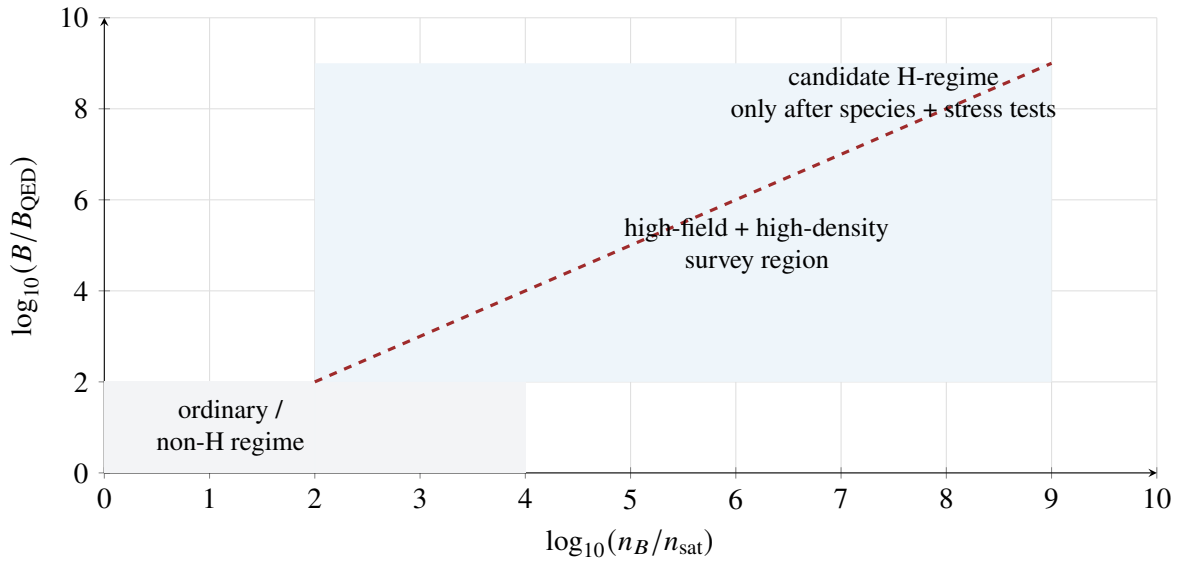


Figure 17.1: Parameter-space survey map. The boundaries are deliberately illustrative: the actual H-regime is defined by the joint diagnostics χ_B, $\nu_{\max,i}$, β_B, density, and the relative electromagnetic/QED contribution to stress.

18
Verification Matrix for a Trustworthy Result

A result should pass multiple independent checks before it is interpreted physically. The following diagram compresses the verification hierarchy into a practical research gate.

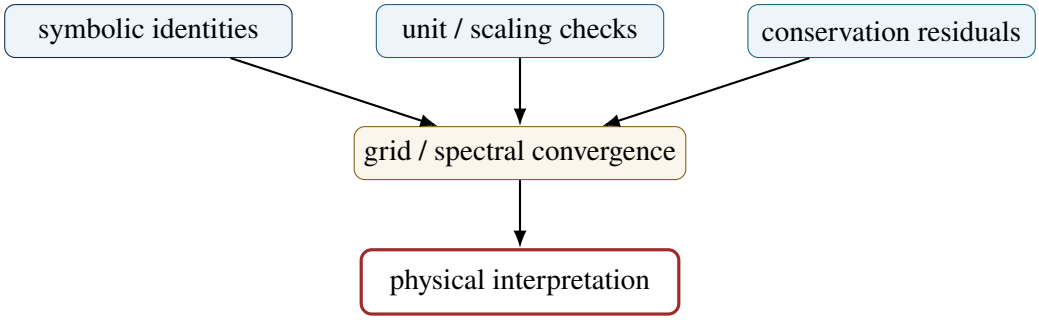


Figure 18.1: Verification gate. A beautiful diagram or a converged number is not sufficient unless the equations, units, conservation laws and numerical errors also close consistently.

19

Research Claim Ladder

The revised theory benefits from a strict hierarchy of claims. This avoids turning a successful numerical effect into an unsupported ontological statement.

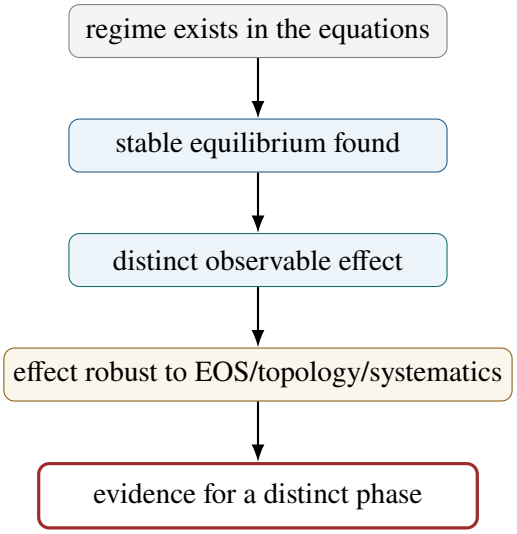


Figure 19.1: Claim ladder. H-Matter begins as a regime label; a stronger phase claim is the final step and requires thermodynamic or order-parameter evidence.

20

Research Evidence Matrix

The following matrix separates the status of each ingredient. This is important because the same physical phenomenon may be well established in one sector and still leave an open HMGQM-specific question in another.

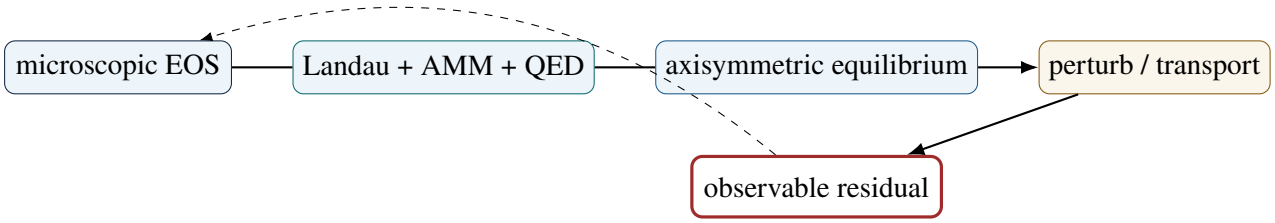
sector	established / benchmarked	HMGQM-specific question
QED	vacuum polarization; finite-field response	does the coupled interior change the observed transport?
dense matter	Landau quantization; AMM; composition	does a coupled solution create a new regime or phase?
GR + EM	axisymmetric equilibrium calculations exist	does the H-regime produce distinct global observables?
observations	magnetar polarization; GW constraints	can residuals survive systematic and EOS uncertainties?

Figure 20.1: Research evidence matrix. The model is strongest when it explicitly separates what has already been demonstrated in the literature from what remains a falsifiable HMGQM question.

21

From Local Physics to an Astrophysical Observable

Every HMGQM prediction passes through several layers. A microscopic change does not become an observation until it survives global equilibrium and transport.



The loop is closed by comparing the residual to a baseline model and propagating EOS, field-topology and transport uncertainties.

Figure 21.1: End-to-end physical hierarchy. The same formal structure appears repeatedly in the theory: local microphysics feeds global geometry, which feeds the observable sector.

22

The Three-Dimensional HMGQM Research Space

The theory naturally spans at least three control directions: magnetic strength, baryon density, and field topology. Composition and temperature add further dimensions. A single one-dimensional scan in B is therefore insufficient to map the model.

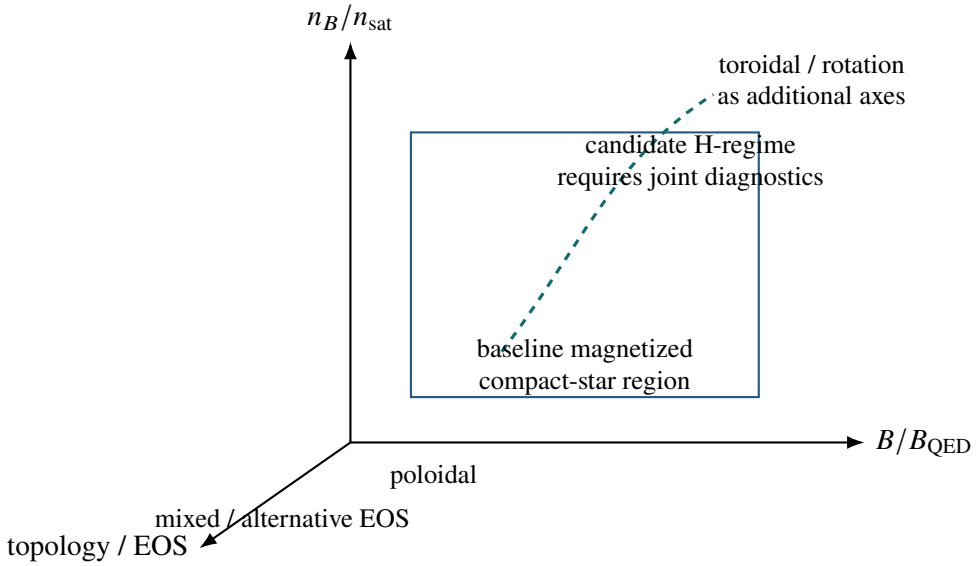


Figure 22.1: Research-space schematic. HMGQM is a multidimensional boundary-value problem; field amplitude alone is not a sufficient coordinate for its physical classification.

23

Research Synthesis: What the Literature Already Tells Us

The surrounding literature narrows the space in which HMGQM can legitimately claim novelty.

23.1 Strong-field QED

The Heisenberg-Euler effective action is a standard low-energy QED tool for nonlinear vacuum response. A finite-field proper-time representation is preferable to a low-field polynomial truncation when the dimensionless field B/B_{QED} is large. The strong-field asymptotic structure is not equivalent to retaining only a first nonlinear F^4 term. [4, 5, 12]

23.2 Magnetized dense matter

The dense-matter literature already contains explicit calculations of Landau quantization, anomalous magnetic moments, changing composition, and magnetic modifications to the EOS. The 2026 review by Sinha and Thapa discusses interior fields as high as 10^{17}-10^{18} G in theoretical studies and emphasizes composition and structural consequences. The 2026 hyperonic analysis by German et al. explicitly compares Landau and AMM effects and finds that their competition can modify stiffness and composition while remaining compatible with approximately $2\,M_{\odot}$ neutron-star observations. [10, 11]

23.3 Relativistic magnetic equilibria

Bocquet et al. demonstrated that magnetized rotating neutron stars can be modeled through fully relativistic, self-consistent Einstein-Maxwell equations and that magnetic deformation becomes important at sufficiently large fields. This establishes the axisymmetric machinery required by HMGQM; it does not by itself establish H-Matter. [7]

23.4 Polarization observations

The 2026 Nature observation of 1E 1547.0-5408 strengthens the empirical motivation for precision magnetar-polarization modeling. HMGQM must therefore aim beyond merely predicting vacuum birefringence: that effect already belongs to standard strong-field QED. The distinctive test would be a statistically and physically robust difference in the observable polarization pattern caused by the HMGQM interior solution after the same exterior transport physics is included. [13, 15]

23.5 Gravitational-wave discipline

GW170817 provides a strong multi-messenger compact-object benchmark, but its post-merger signal was not detected. HMGQM should therefore use future detected events, inspiral tidal constraints, continuous-wave limits, or dedicated post-merger searches as prospective tests rather than retrofitting an undetected frequency into an observational plot. [14, 18]

Research conclusion

The literature does not remove the motivation for HMGQM. It changes the standard of proof. The theory is scientifically interesting only if its coupled, axisymmetric, finite-field calculation yields consequences that survive comparison against established magnetized compact-star models and their uncertainties.

Part III

Research-Informed Expansion and Diagram Atlas

24

Scale Separation: Microphysics, Macrophysics, and the H-Regime

A central lesson from the surrounding literature is that magnetic strength, density, curvature, composition, and global field topology are different control variables. HMGQM therefore should not be reduced to a single threshold such as $B > B_{\mathrm{QED}}$. The 2026 dense-matter review surveys Landau quantization, anomalous magnetic moments, composition changes, and structural effects in strong fields, while fully relativistic magnetic-star calculations demonstrate that the global geometry must be solved self-consistently. [10, 7, 8]

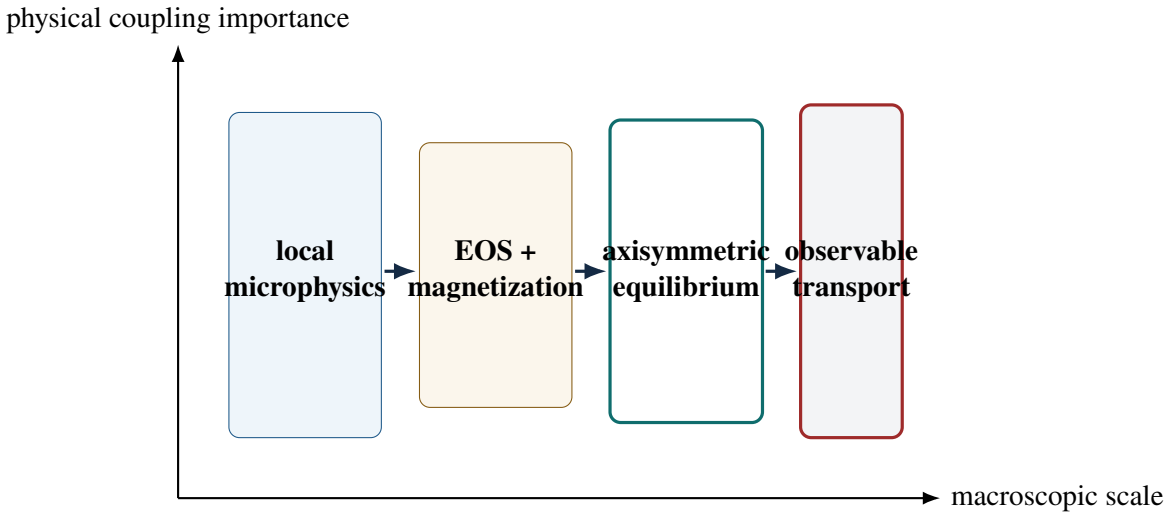


Figure 24.1: Scale hierarchy for the HMGQM program. Each layer has its own equations and uncertainties; the proposed theory becomes predictive only when information is passed consistently between layers.

25

Explicit Landau Spectrum and Level Occupation

For a charged species i with charge q_i and effective mass m_i^*, a minimal local spectrum is

$$E_{i\nu}^2 = p_z^2 + (m_i^*)^2 + 2\nu|q_i|B, \qquad \nu = 0, 1, 2, \ldots \tag{25.1}$$

with $g_\nu = 2 - \delta_{\nu 0}$. The occupied-level count is therefore a dynamical output of the chemical potential, interactions, and field strength. [16, 11, 10]

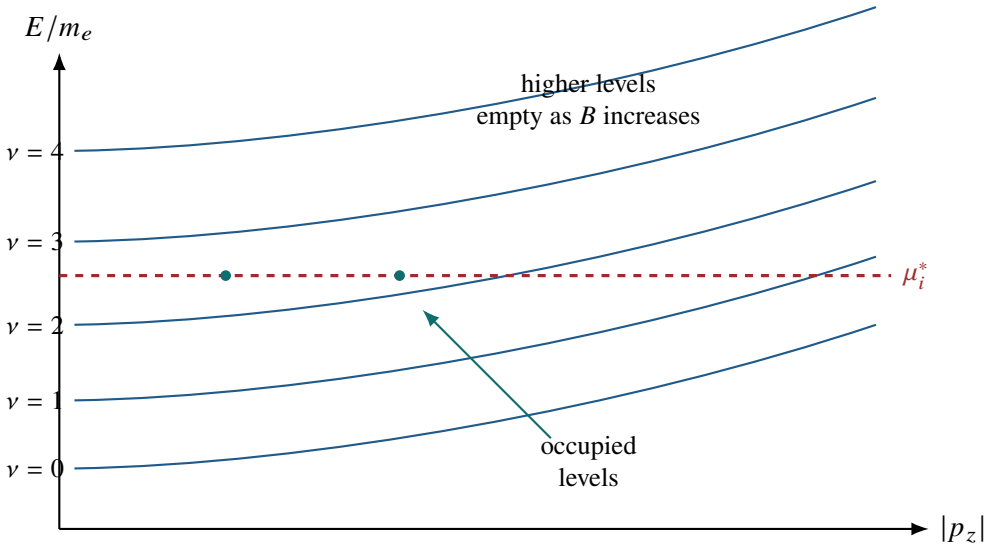


Figure 25.1: Landau-level fan schematic. The Fermi surface intersects only those levels with real longitudinal Fermi momentum. Stronger B increases the spacing between levels, but the lowest-level regime must still be checked species by species.

26

Which Species Feel the Field First?

The electron sector is the cleanest entry point because B_{QED} is defined using m_e, whereas the much heavier baryons have very different characteristic scales. This is why the HMGQM criterion uses a species-resolved $\nu_{\max,i}$ instead of declaring all matter one-dimensional. [16, 11]

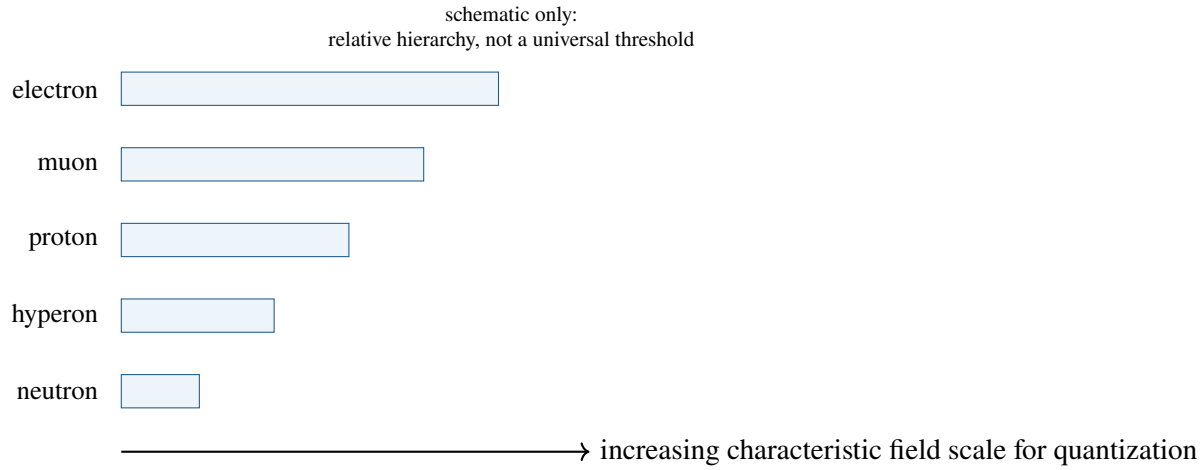


Figure 26.1: Species sensitivity hierarchy. The figure is deliberately schematic: exact thresholds depend on chemical potentials, interactions, anomalous magnetic moments, and composition. Neutrons are neutral and are not Landau quantized, although their magnetic moments can still couple to the field.

27

Magnetic Stress versus Matter Stiffness

The magnetic contribution is not merely a microscopic correction. In the total stress budget, it can become comparable with the material pressure. The 1995 and 2001 relativistic calculations already show that sufficiently strong fields deform compact stars, while the 2026 literature emphasizes that Landau quantization and AMM effects can alter the matter sector simultaneously. [7, 8, 10, 11]

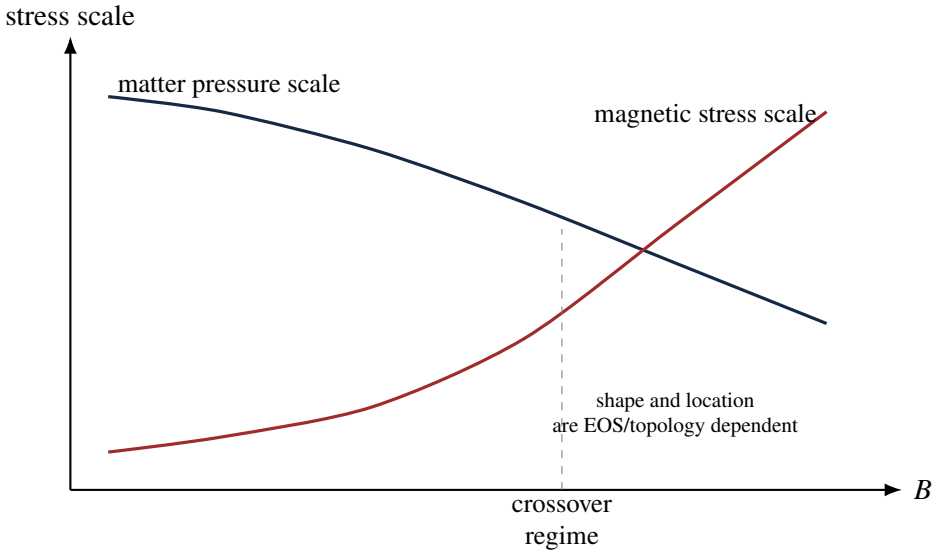


Figure 27.1: Stress-budget schematic. A crossover between matter and magnetic stresses is not itself a phase transition; it identifies a regime in which magnetic geometry can no longer be treated as a perturbation of an isotropic star.

28

Grand Potential to Observable Stress

The revised framework uses the thermodynamic potential as the local closure. From a renormalized grand potential Ω, magnetization is $M = -\partial\Omega/\partial B$ at fixed thermodynamic control variables, and the principal stresses follow from the chosen convention for separating material and field contributions. The important methodological point is consistency: the same convention must be used in the action, thermodynamics, and relativistic stress tensor. [6, 16]

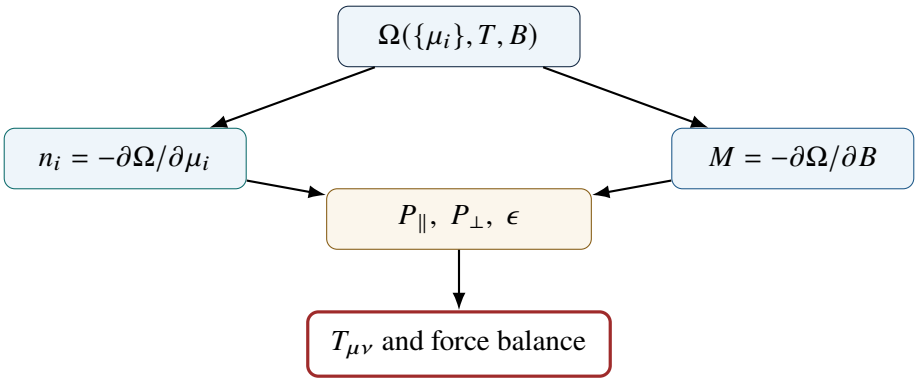


Figure 28.1: Thermodynamic Legendre map. The grand potential is not an ornament: it is the bridge from microscopic composition to a consistent stress-energy closure.

29

Stress Tensor Geometry in the Local Rest Frame

The most compact covariant form is

$$T^{\mu\nu} = \epsilon u^\mu u^\nu + P_\perp h^{\mu\nu} + (P_\parallel - P_\perp) b^\mu b^\nu, \tag{29.1}$$

with $h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$, $b^\mu b_\mu = 1$, and $u^\mu b_\mu = 0$. This representation makes the anisotropy geometric rather than a coordinate artifact. [2]

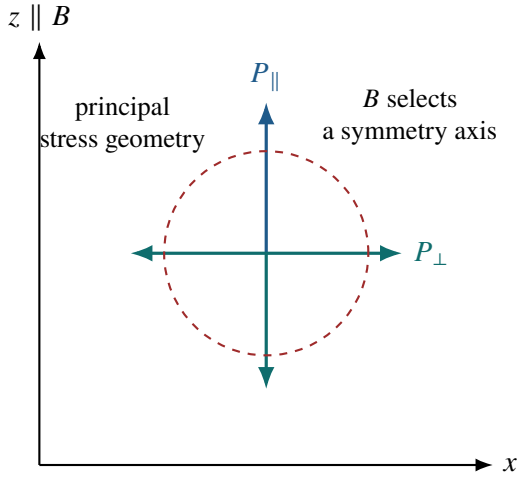


Figure 29.1: Local stress geometry. When $P_\parallel \neq P_\perp$, the material sector cannot be represented by a single scalar pressure.

30

Poloidal Magnetic Topology

The global magnetic field is not specified by its local magnitude. A poloidal component threads meridional planes and closes through the exterior magnetosphere. Fully relativistic axisymmetric calculations use a magnetic four-potential and field geometry as part of the boundary-value problem. [7]

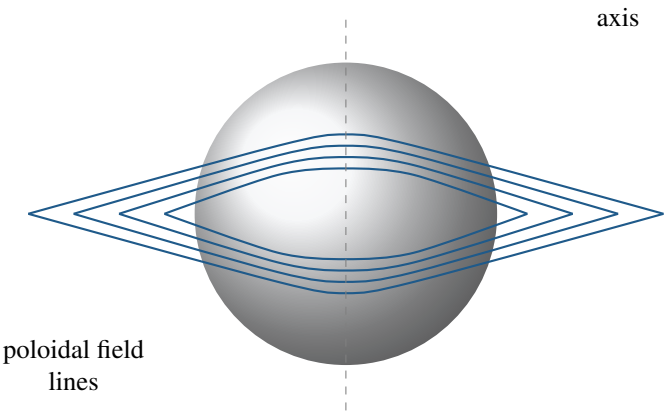


Figure 30.1: Poloidal field topology. The field lines thread the object and extend into the exterior. The exact topology is determined by the current distribution and boundary conditions, not by the H-Matter label alone.

31

Toroidal Magnetic Topology

Toroidal magnetic energy can make a substantial contribution to deformation, but it is not interchangeable with the poloidal sector. Any claim about an HMGQM configuration must therefore specify which topology is being solved. [17, 10]

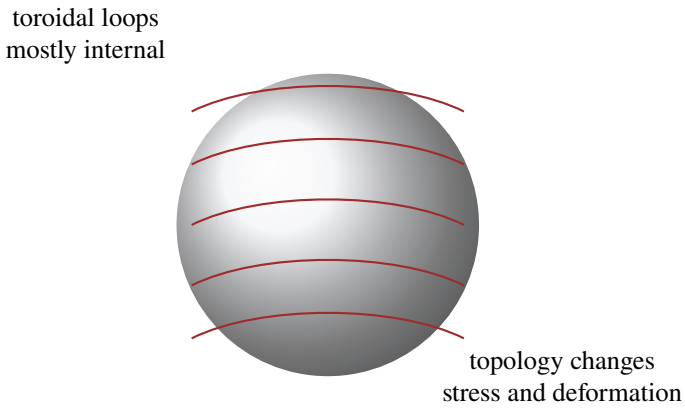


Figure 31.1: Toroidal-field schematic. Toroidal configurations are a distinct magnetic geometry and can redistribute magnetic energy and deformation differently from poloidal fields.

32

Mixed Poloidal-Toroidal Configuration

A mixed configuration provides a more general test of robustness. The same central field amplitude can correspond to different ellipticities, energy partitions, and exterior signatures depending on topology. [17]

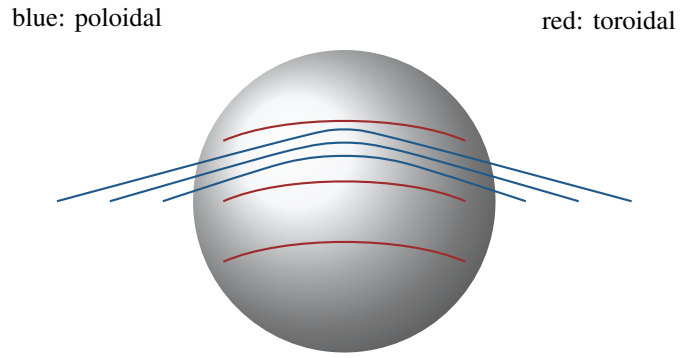


Figure 32.1: Mixed topology. HMGQM should ultimately be tested across a family of magnetic geometries because topology is part of the physical state, not just a visualization choice.

33

Axisymmetric Star Cross-Section

In the global problem, pressure anisotropy and electromagnetic stress deform the matter distribution while the magnetic field reshapes the gravitational source. The stationary axisymmetric metric used in v2.0 and v3.1 captures this geometry. [2, 7]

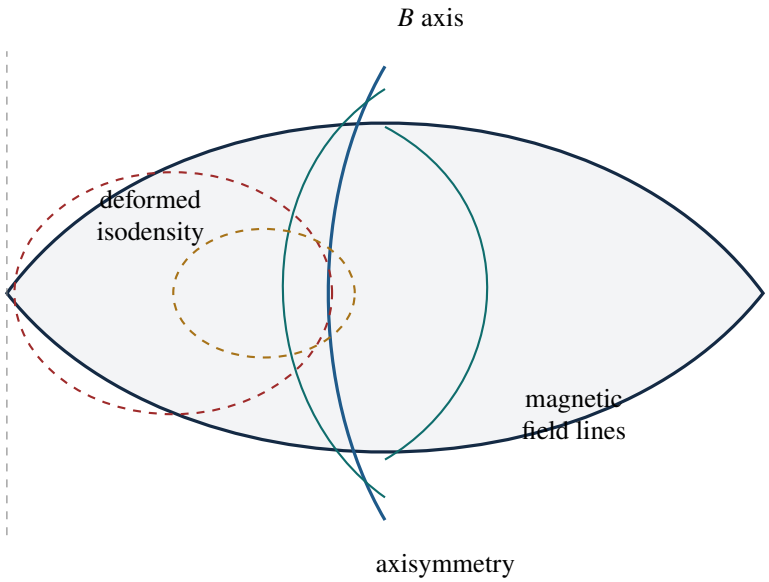


Figure 33.1: Conceptual meridional section of an axisymmetric compact object. Isodensity contours, field lines, and the symmetry axis form a coupled geometry.

34

Surface Matching and the Exterior Magnetosphere

A physically complete model must match the interior solution to an exterior electrovacuum or magnetosphere model. Observed polarization and spectra are formed along this exterior path; they cannot be inferred from an interior stress tensor alone. [7, 13]

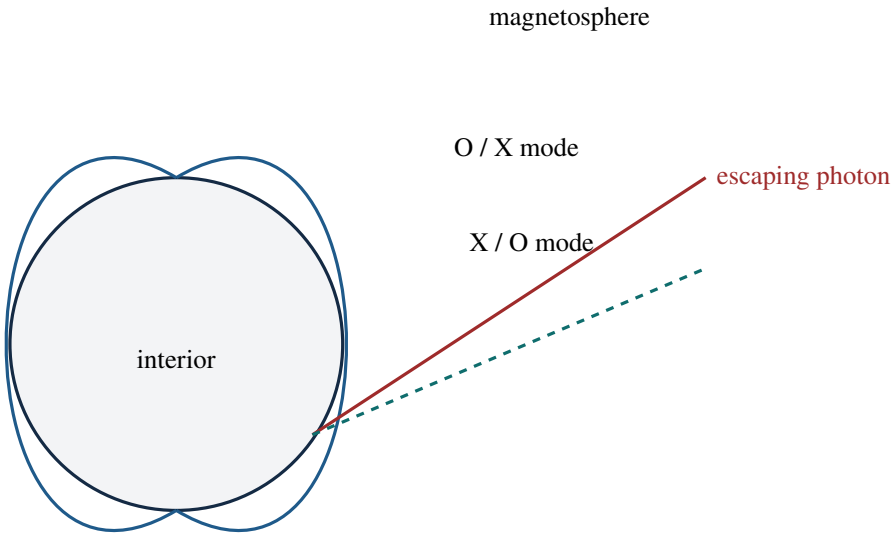


Figure 34.1: Interior-to-exterior transport chain. Two photon polarization modes can follow different effective propagation laws in strong magnetic fields. A HMGQM observable must include the exterior transport and surface emission model consistently. [13, 12]

35

Finite-Field Birefringence as a High-Value Test Channel

The 2026 magnetar observation of 1E 1547.0-5408 reported large X-ray polarization, including phase-averaged polarization of 65% at 2 keV and values approaching 80% at selected rotational phases. The observation supports the need for QED-consistent propagation modeling but does not uniquely establish HMGQM, because vacuum birefringence is already part of standard QED. [13]

A 2026 finite-field calculation further reported that a weak-field Cotton-Mouton approximation can overestimate the accumulated birefringent phase by up to a factor 2.9 at 10^{15} G and identified vacuum-resonance observables as particularly sensitive to finite-field corrections. These are useful targets for HMGQM because the theory explicitly retains a finite-field one-loop sector. [12]

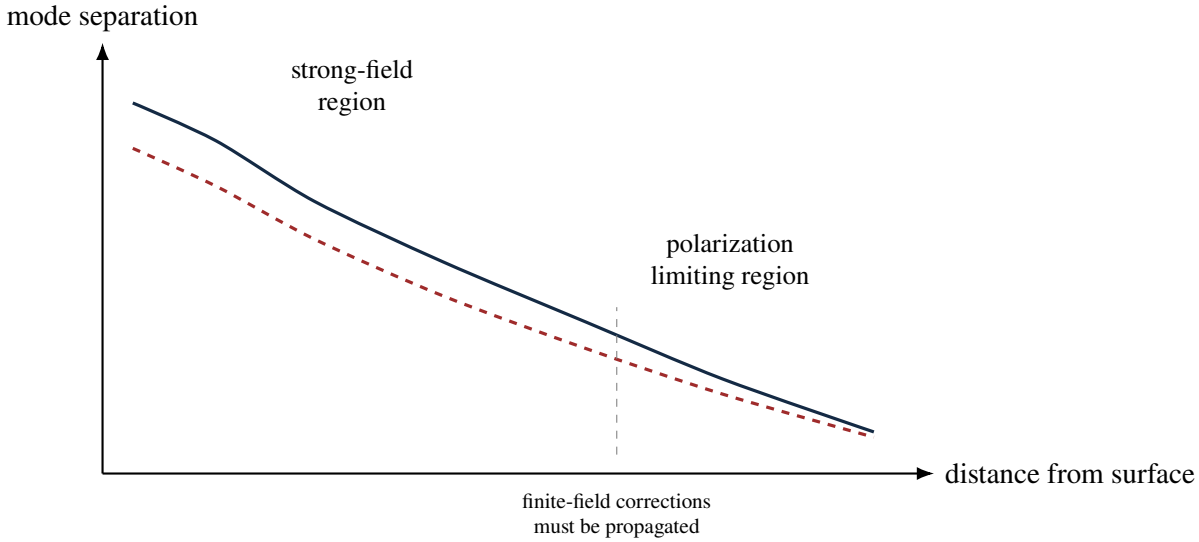


Figure 35.1: Schematic polarization transport hierarchy. The near-surface region can be the most sensitive to finite-field response, while mode decoupling can occur farther out where the local field has weakened substantially. The exact curves are model dependent. [12]

36

Gravitational-Wave Mode Accounting

The original draft associated GW170817 and GW190814 with points on an oscillation-frequency plot. The revised framework removes that interpretation because GW170817 post-merger oscillation was not directly detected. A correct HMGQM calculation should instead predict a mode spectrum from the linearized Einstein-Maxwell-matter system and compare measurable mode content with a baseline. [14, 18]

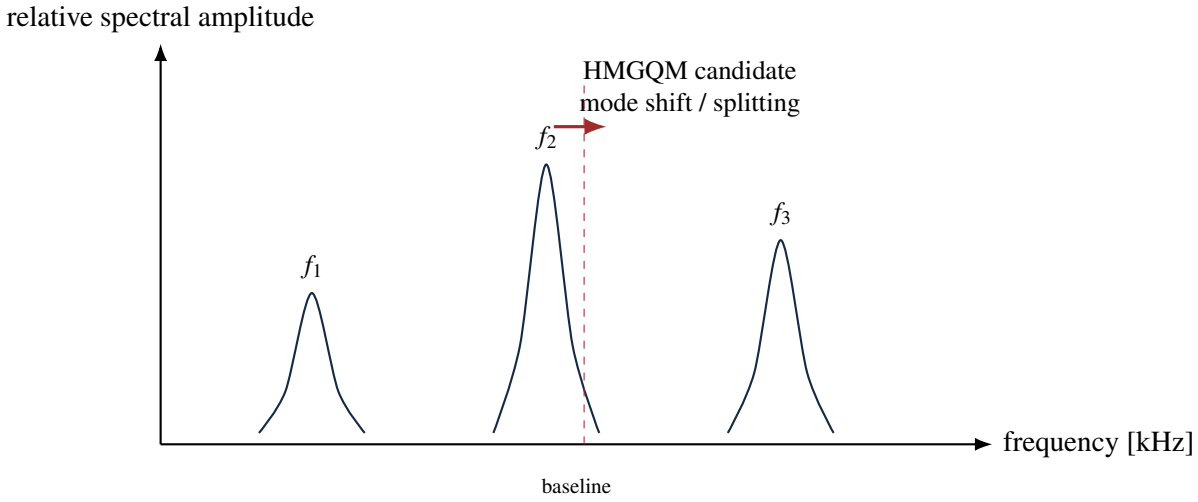


Figure 36.1: Mode-spectrum schematic. In an axisymmetric magnetized configuration, degeneracies and mode frequencies can be shifted or split. The numerical shift is a prediction only after the coupled perturbation equations are solved.

37

Tidal and Rotational Diagnostics

A robust test should use more than one scalar observable. Current EOS studies emphasize complementary constraints from high-mass stars, radius measurements, and tidal deformability. The HMGQM program can therefore report $M(R)$, $\Lambda(M)$, moment of inertia, magnetic ellipticity, and rotational response from the same equilibrium sequence. [20]

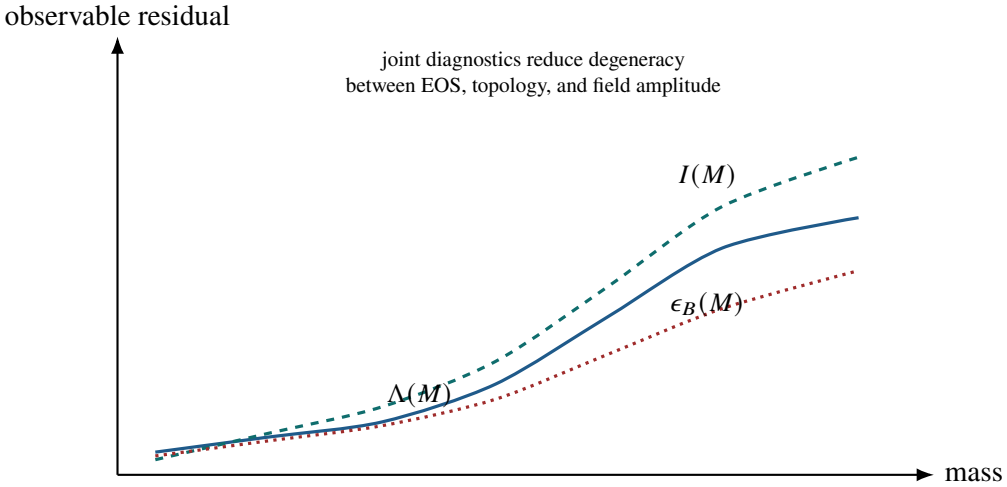


Figure 37.1: Multi-observable diagnostic strategy. HMGQM should be evaluated as a vector of residuals rather than a single curve.

38

Parameter Degeneracy and Model Comparison

A difference between HMGQM and a baseline model is meaningful only after nuisance parameters are propagated. EOS stiffness, field topology, surface temperature, inclination, distance, and radiative-transfer assumptions can all move an inferred observable. The correct question is therefore whether an HMGQM signature survives the full uncertainty volume. [20, 13]

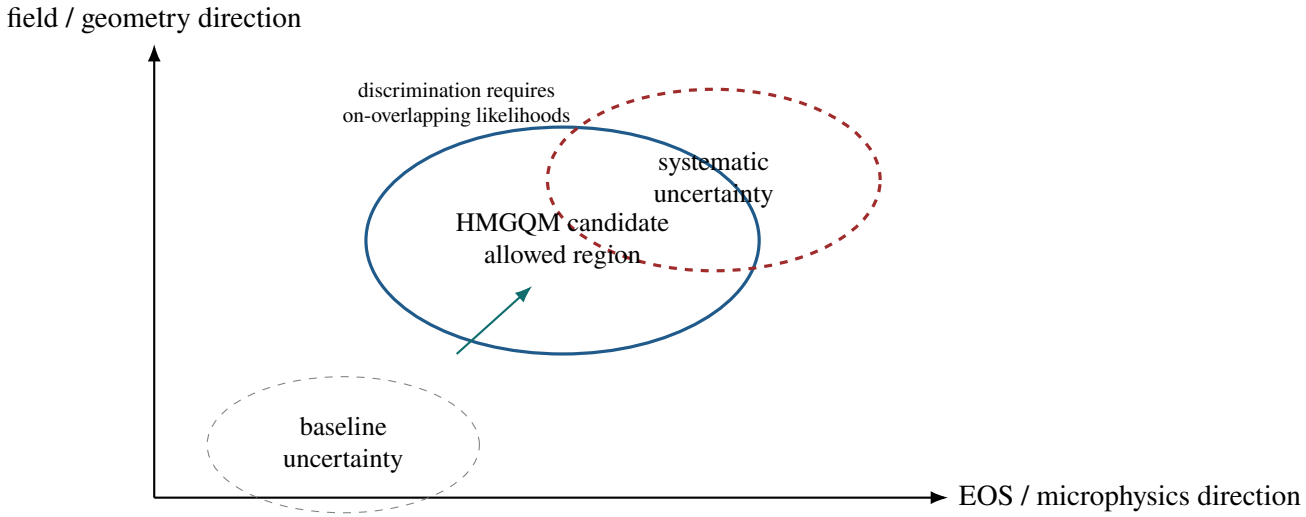


Figure 38.1: Parameter-degeneracy schematic. An apparent HMGQM residual should not be interpreted as evidence for a new phase when it can be absorbed by plausible EOS or geometry uncertainty.

39

Numerical Convergence as a Scientific Requirement

A numerical equilibrium is not a result until convergence is demonstrated. For a sequence of grid or spectral resolutions N, a representative diagnostic should approach a limiting value with a controlled error estimate. The exact order depends on the numerical method. [7]

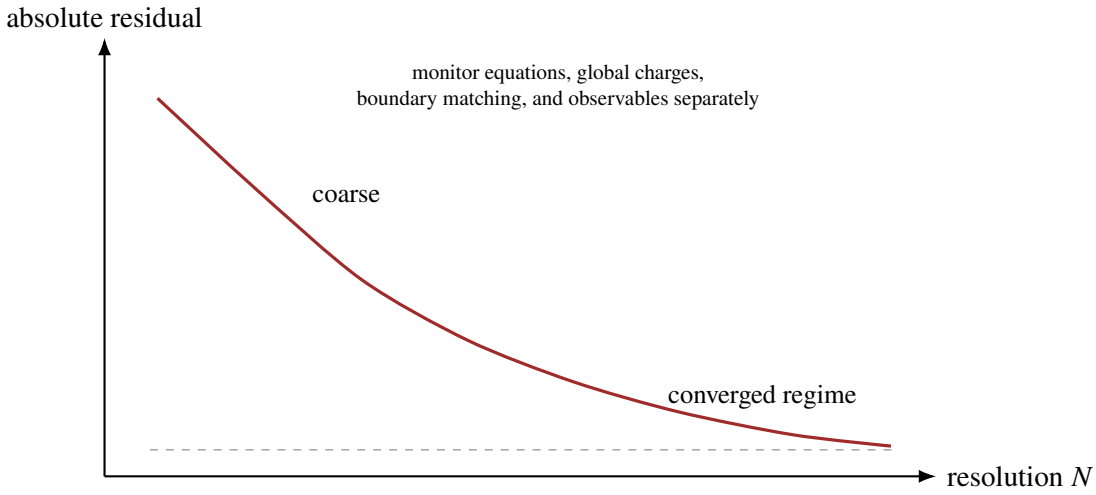


Figure 39.1: Convergence schematic. A visually smooth solution is not enough; residuals of the field equations and conservation laws must also decrease under refinement.

40

Uncertainty Propagation Pipeline

HMGQM has uncertainty at several layers: microphysics, magnetic topology, numerical discretization, exterior transport, and measurement. A publishable prediction should carry these uncertainties through to a final posterior or confidence interval rather than quoting a single deterministic curve. [20]

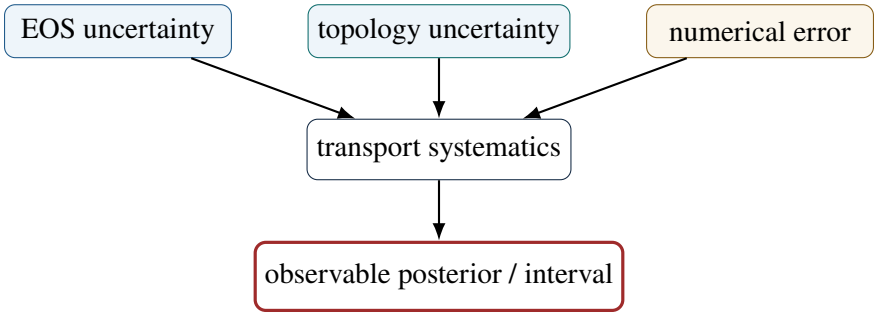


Figure 40.1: Uncertainty propagation pipeline. The HMGQM hypothesis is tested after nuisance uncertainties are propagated, not before.

41

Regime Label versus Phase Discovery

The word H-Matter has two possible meanings and should not be allowed to blur them. The first is an operational regime in which high field, high density, Landau structure and magnetic/QED stress are jointly important. The second is a claim of a distinct thermodynamic phase, which requires an order parameter or non-analytic thermodynamic feature. [2]

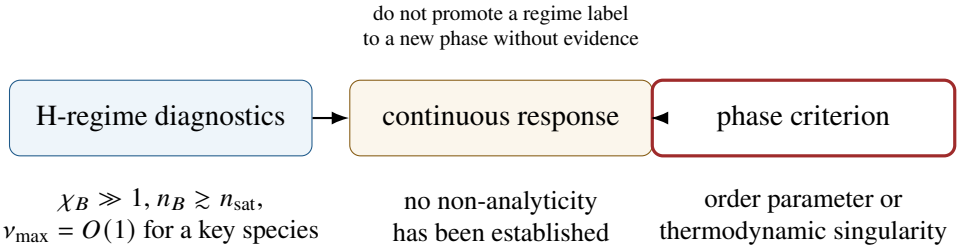


Figure 41.1: Scientific decision boundary between an extreme regime and a newly established phase. This distinction is one of the most important safeguards in the revised theory.

42

The Evidence Ladder

A mature HMGQM program should move through increasingly demanding evidence levels: analytic consistency, numerical convergence, agreement with known limits, comparison to established magnetized-star models, and only then observational discrimination. This structure prevents a suggestive diagram from being mistaken for a confirmed physical effect.

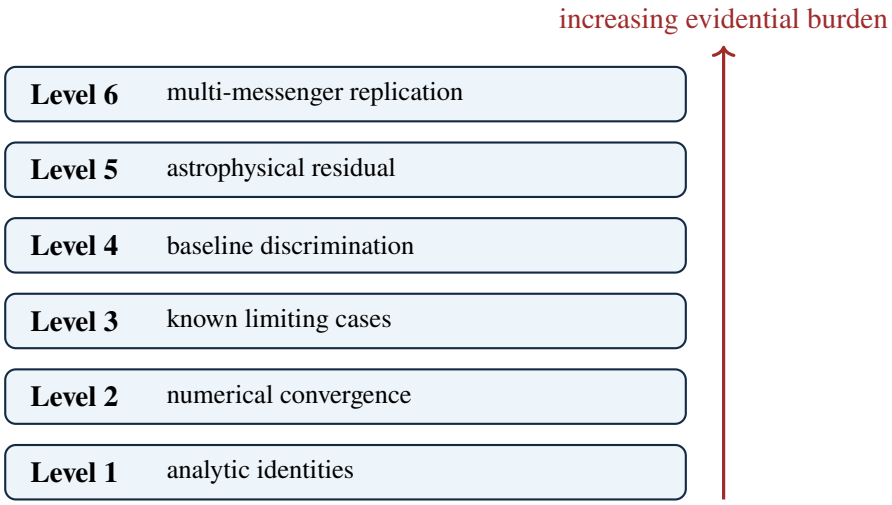


Figure 42.1: Evidence ladder. HMGQM should be considered scientifically stronger as it climbs from formal consistency to independently reproducible observation.

43

Falsification Decision Tree

A theory is most useful when it specifies how it can fail. The HMGQM implementation should be rejected or narrowed if it violates conservation, fails the known limits, cannot achieve numerical convergence, produces unstable equilibria where stability is claimed, or offers no observable residual after uncertainty propagation.

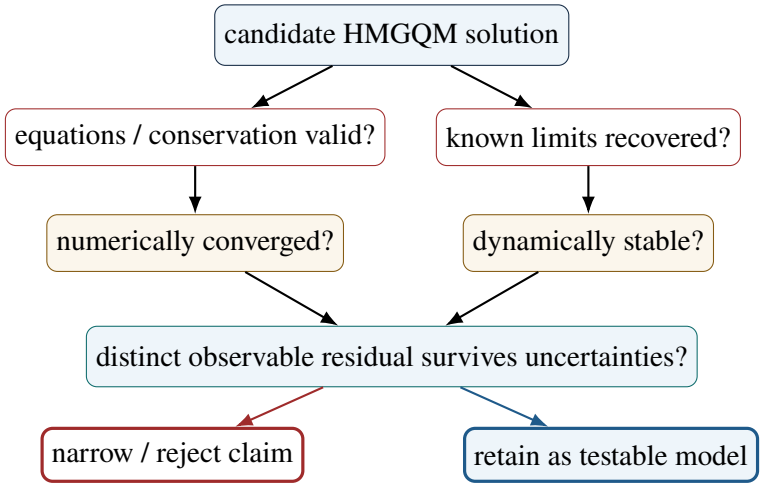


Figure 43.1: Falsification decision tree. A candidate must pass mathematical consistency, known-limit recovery, numerical convergence, and dynamical-stability checks before any distinctive observational residual can support a stronger claim.

44

Reproducibility and Independent Reimplementation

The framework is modular enough that two implementations can disagree while both appearing plausible. A reproducibility package should therefore specify constants, EOS table or Lagrangian, unit convention, magnetic topology, grid or spectral basis, solver tolerances, surface matching, transport model, and all derived observables. The literature on relativistic magnetic equilibria illustrates the importance of self-consistent numerical tests. [7]

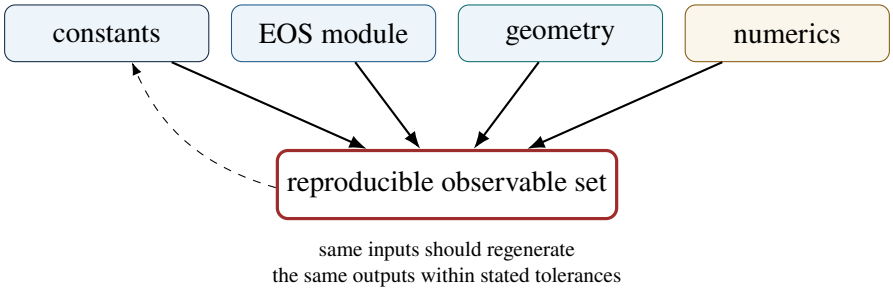


Figure 44.1: Reproducibility manifest. The feedback arrow represents versioned inputs and preserved numerical diagnostics, enabling independent reimplementation.

45

Research Update: What Changed the Standard of Proof

The most important research-driven changes in this edition are methodological rather than rhetorical. Strong-field QED calculations motivate keeping the finite-field response instead of extrapolating a truncated low-field series. Dense-matter studies motivate species-resolved Landau and AMM treatments. Relativistic magnetic-star calculations motivate a genuine axisymmetric Einstein-Maxwell equilibrium solver. Recent magnetar polarization observations motivate a transport comparison in which standard QED birefringence is treated as a baseline rather than an HMGQM-exclusive signature. [12, 10, 11, 7, 13]

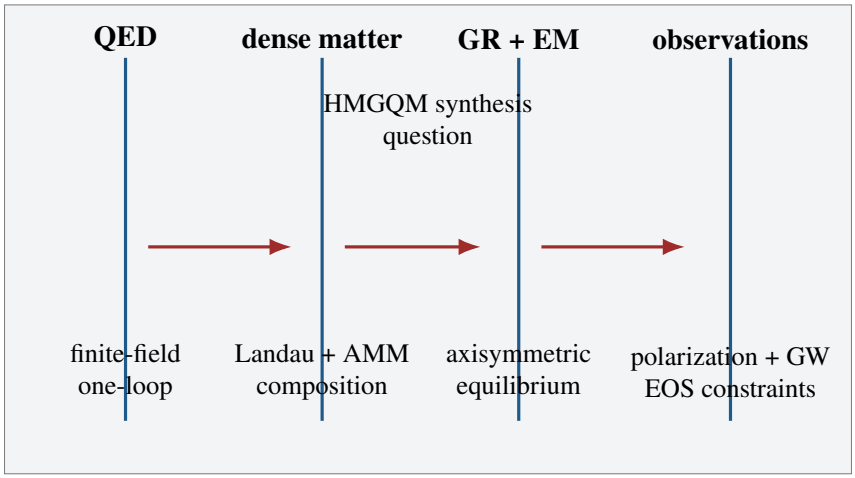


Figure 45.1: Research synthesis map. The literature supplies the ingredients and constraints; HMGQM is a hypothesis about their coupled extreme-regime consequences.

Part IV

Covariant Mathematical Formulation

46

Units, Conventions, and Variational Structure

46.1 Natural-unit convention

The derivation is performed in Heaviside-Lorentz natural units, $\hbar = c = k_B = \epsilon_0 = \mu_0 = 1$, with metric signature $(-,+,+,+)$. The electromagnetic invariants are

$$\mathcal{F} = \frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \qquad \mathcal{G} = \frac{1}{4} F_{\mu\nu} \tilde{F}^{\mu\nu}. \tag{46.1}$$

For a purely magnetic field in the comoving frame, $\mathcal{F} = B^2/2$ and $\mathcal{G} = 0$.

46.2 The defining action

The complete effective action is

$$\boxed{S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \Delta\mathcal{L}_{\rm HE}(\mathcal{F}, \mathcal{G}) + \mathcal{L}_{\rm matter} \right].} \tag{46.2}$$

Here $\mathcal{L}_{\rm matter}$ denotes the explicitly chosen interacting dense-matter microphysics. The action is intentionally modular: the HMGQM framework specifies the coupling structure, while the microphysical closure can be implemented through a relativistic mean-field model, chiral EFT matched to a high-density closure, a tabulated microscopic EOS, or a controlled quark-matter model.

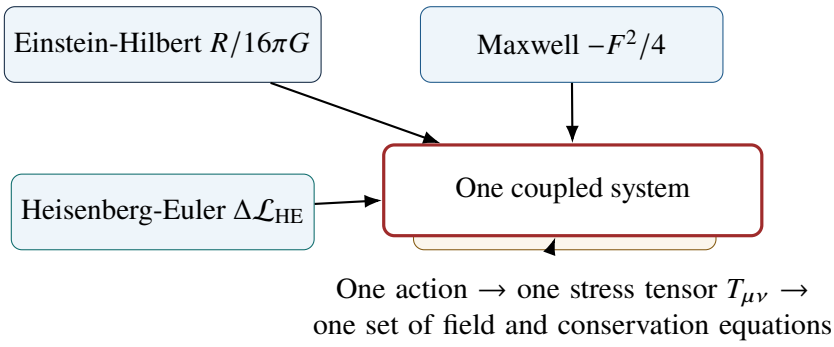


Figure 46.1: Action architecture. The diagram is the central anti-double-counting rule of the theory.

47

Strong-Field QED: Finite-Field Heisenberg-Euler Sector

47.1 Why the weak-field truncation is not the defining model

A common low-field representation expands the QED effective action as a power series in B/B_{QED}. Such a Taylor expansion is useful as a consistency check for $B \ll B_{\mathrm{QED}}$, but it cannot be used as the defining response when the theory explicitly targets $B \gg B_{\mathrm{QED}}$. [4, 5]

47.2 Renormalized proper-time representation

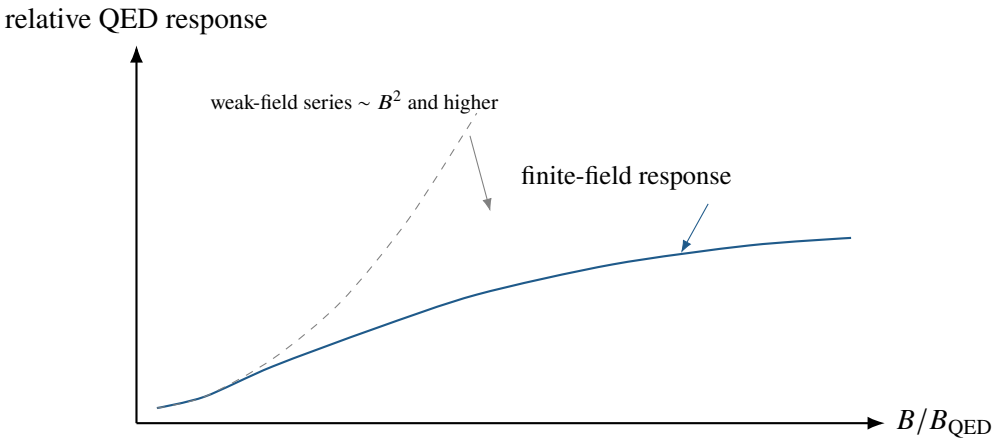
For a slowly varying, purely magnetic background and photon energies below the pair-production threshold, a standard one-loop spinor-QED representation is

$$\boxed{\Delta\mathcal{L}_{\mathrm{HE}}(B) = -\frac{1}{8\pi^2}\int_0^\infty \frac{ds}{s^3} e^{-m_e^2 s}\left[eBs\coth(eBs) - 1 - \frac{(eBs)^2}{3}\right],} \tag{47.1}$$

with the renormalized $B = 0$ vacuum piece and charge-renormalization subtraction understood. This form retains the non-polynomial dependence on B that is essential to the strong-field problem. [4]

47.3 Asymptotic strong-field behavior

The strong-field limit contains logarithmic behavior and, beyond one loop, additional reducible contributions. The theory therefore keeps the one-loop finite-field functional as its minimal QED sector and treats higher-loop corrections as an uncertainty to be bounded rather than silently neglected. [5]



Schematic only: finite-field response is retained; the dashed low-field series is shown only to illustrate why asymptotic expansions should not be extrapolated indefinitely.

Figure 47.1: Finite-field versus truncated strong-field response. The plot is schematic and is not a numerical fit.

47.4 Constitutive tensor

Define

$$H^{\mu\nu} \equiv -2\frac{\partial(\mathcal{L}_{\text{EM}} + \Delta\mathcal{L}_{\text{HE}})}{\partial F_{\mu\nu}}. \tag{47.2}$$

Then

$$\nabla_\mu H^{\mu\nu} = J^\nu, \qquad \nabla_{[\alpha}F_{\beta\gamma]} = 0. \tag{47.3}$$

The second relation is the Bianchi identity and implies $\nabla_\mu \tilde{F}^{\mu\nu} = 0$.

48

Einstein Equation and Stress-Energy Accounting

Metric variation of the full action gives

$$G_{\mu\nu} = 8\pi G\, T_{\mu\nu}, \tag{48.1}$$

with

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_{\text{non-grav}}}{\delta g^{\mu\nu}}. \tag{48.2}$$

The total tensor is therefore a single object containing matter, Maxwell, and QED-vacuum response.

48.1 Local anisotropic decomposition

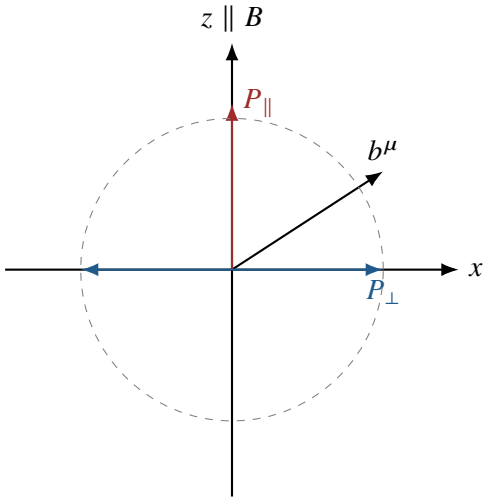
In the local rest frame of the matter, define $u_\mu u^\mu = -1$, $b_\mu b^\mu = 1$, and $u_\mu b^\mu = 0$. Let

$$h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu. \tag{48.3}$$

Then the total local stress tensor can be written

$$\boxed{T_{\mu\nu} = \epsilon u_\mu u_\nu + P_\perp h_{\mu\nu} + (P_\parallel - P_\perp) b_\mu b_\nu.} \tag{48.4}$$

This is a geometric statement about principal stresses; it is not a declaration that two arbitrary scalar pressures can be inserted independently.



The magnetic axis selects one principal direction. In an isotropic limit $P_\parallel = P_\perp = P$.

Figure 48.1: Local principal-stress geometry.

48.2 Conservation law

Diffeomorphism invariance yields

$$\boxed{\nabla_\mu T^{\mu\nu} = 0.} \tag{48.5}$$

This equation is not optional. In numerical work, failure to satisfy it beyond the discretization tolerance is a consistency failure, not merely a small plotting error.

Part V

Microscopic Dense Matter and Magnetized Thermodynamics

49
Particle Content and Chemical Equilibrium

49.1 Minimal reference composition

The baseline cold matter model contains neutrons, protons, electrons, and, when energetically allowed, muons. More elaborate calculations can add the baryon octet, resonances, condensates, or quarks. [10, 11]

For cold charge-neutral npeμ matter,

$$n_p = n_e + n_\mu, \tag{49.1}$$

and beta equilibrium gives

$$\mu_n - \mu_p = \mu_e = \mu_\mu. \tag{49.2}$$

These are closure conditions rather than independent HMGQM inventions.

49.2 Why the composition matters

The field changes the density of states of charged species, which shifts chemical potentials at fixed total baryon density. A self-consistent model must therefore solve field-dependent composition rather than applying a magnetic correction after the fact.

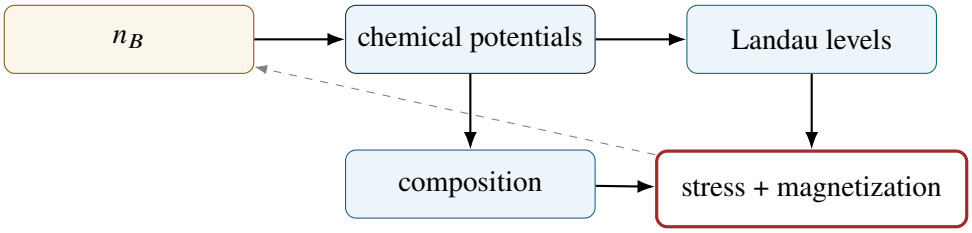


Figure 49.1: Self-consistent composition loop. The dashed arrow represents feedback through equilibrium and the EOS.

50

Landau Quantization

50.1 Charged relativistic fermions

For a charged fermion of charge q_i and effective mass m_i^* in a locally uniform magnetic field,

$$E_{i\nu}(p_z) = \sqrt{p_z^2 + (m_i^*)^2 + 2\nu|q_i|B}, \qquad \nu = 0, 1, 2, \ldots \tag{50.1}$$

with spin degeneracy

$$g_\nu = 2 - \delta_{\nu 0}. \tag{50.2}$$

At $T = 0$ the Fermi momentum at each level is

$$k_{F,i\nu} = \sqrt{(\mu_i^*)^2 - (m_i^*)^2 - 2\nu|q_i|B}, \tag{50.3}$$

provided the radicand is non-negative. Thus

$$\nu_{\max,i} = \left\lfloor \frac{(\mu_i^*)^2 - (m_i^*)^2}{2|q_i|B} \right\rfloor. \tag{50.4}$$

The density becomes

$$\boxed{n_i = \frac{|q_i|B}{2\pi^2} \sum_{\nu=0}^{\nu_{\max,i}} g_\nu k_{F,i\nu}.} \tag{50.5}$$

50.2 Zero-temperature grand potential

For the free charged species contribution,

$$\Omega_i = -\frac{|q_i|B}{4\pi^2} \sum_{\nu=0}^{\nu_{\max,i}} g_\nu \left[\mu_i^* k_{F,i\nu} - m_{i\nu}^2 \ln\left(\frac{\mu_i^* + k_{F,i\nu}}{m_{i\nu}} \right) \right], \tag{50.6}$$

where $m_{i\nu}^2 = (m_i^*)^2 + 2\nu|q_i|B$. [6]

50.3 No automatic one-dimensional collapse

Only when $\nu_{\max,i} = 0$ does species i occupy the lowest Landau level. Even when electrons reach that regime, neutrons do not become Landau-quantized because they are electrically neutral. Charged baryons and leptons enter strong quantization at different field strengths because their masses, chemical potentials, and interactions differ. [10, 11]

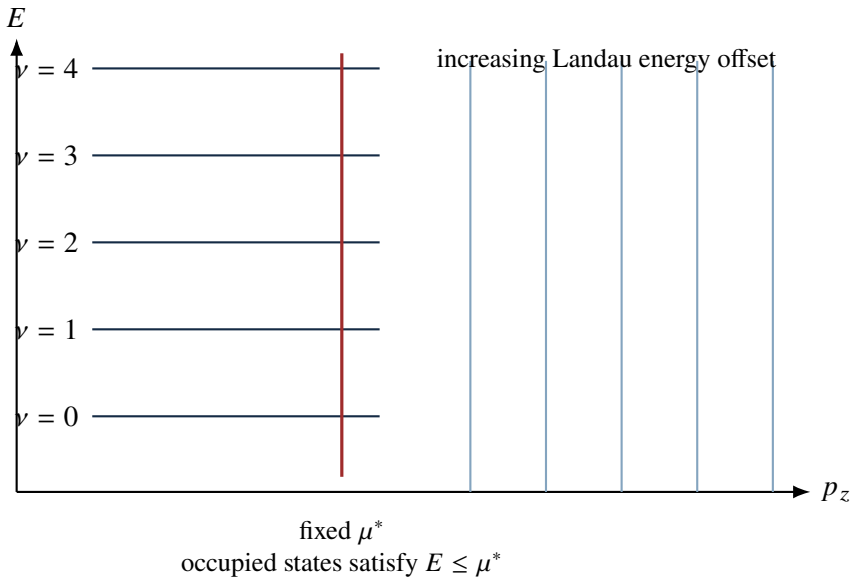


Figure 50.1: Landau-level ladder. The schematic emphasizes that strong fields discretize transverse motion while leaving a continuous longitudinal momentum p_z.

51

Neutral Species and Anomalous Magnetic Moments

Neutral fermions can couple to magnetic fields through anomalous magnetic moments. Charged species can also receive anomalous-magnetic-moment (AMM) corrections in addition to Landau quantization. At very strong fields, AMM-induced spin splitting can compete with Landau quantization and shift the composition and stiffness of matter. Recent hyperonic calculations emphasize that these effects can push the EOS in different directions depending on the species and field profile. [11]

A general interacting closure is therefore written as

$$\Omega_m = \Omega_m(\{\mu_i\}, T, B; \theta_{\mathrm{micro}}), \tag{51.1}$$

where θ_{micro} denotes the stated parameters of the nuclear, hyperonic, or quark model.

52

Thermodynamic Potentials, Magnetization, and Principal Stresses

52.1 Grand potential

Let

$$\Omega = \Omega_m + \Omega_{\mathrm{HE}}^{\mathrm{ren}}. \tag{52.1}$$

The material pressure is $P_m = -\Omega$. The magnetization is

$$\boxed{M = -\left(\frac{\partial \Omega}{\partial B}\right)_{\{\mu_i\},T}.} \tag{52.2}$$

52.2 Energy density

In natural units, the total local energy density can be expressed as

$$\epsilon = \Omega + \sum_i \mu_i n_i + \frac{B^2}{2}, \tag{52.3}$$

with the precise decomposition adjusted consistently to the chosen ensemble and inclusion of vacuum terms.

52.3 Principal pressures

A consistent total-pressure convention is

$$\boxed{P_{\parallel} = -\Omega - \frac{B^2}{2},} \tag{52.4}$$

$$\boxed{P_{\perp} = -\Omega - MB + \frac{B^2}{2}.} \tag{52.5}$$

Therefore

$$\boxed{\Delta P \equiv P_{\perp} - P_{\parallel} = B^2 - MB.} \tag{52.6}$$

These expressions show directly why magnetization is not a decorative correction: M changes the stress anisotropy.

52.4 A warning about signs and conventions

The individual separation of "matter pressure", "magnetic pressure", and "vacuum pressure" depends on ensemble and conventions. The invariant statement is that the total stress tensor follows from one variational definition. Equations above are the adopted natural-unit convention for this document; any numerical code must keep the same convention consistently.

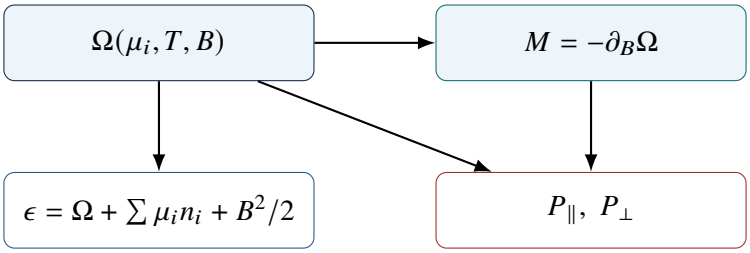


Figure 52.1: Thermodynamic map: one grand potential generates energy, magnetization, and principal stresses.

53

H-Matter as a Regime and as a Possible Phase

53.1 Regime definition

A region is labeled H-Matter when the H-regime inequalities are simultaneously satisfied and magnetic/QED terms make an order-unity contribution to the local or global stress balance.

53.2 Phase criterion

Anisotropy alone does not prove a thermodynamic phase. A stronger phase claim requires a non-analyticity in a thermodynamic potential or a well-defined order parameter Φ. For example,

$$\Phi = 0 \quad \text{outside the phase,} \qquad \Phi \neq 0 \quad \text{inside,} \tag{53.1}$$

with a discontinuity, divergent susceptibility, or other non-analytic signature at the boundary, appropriate to the order of the transition.

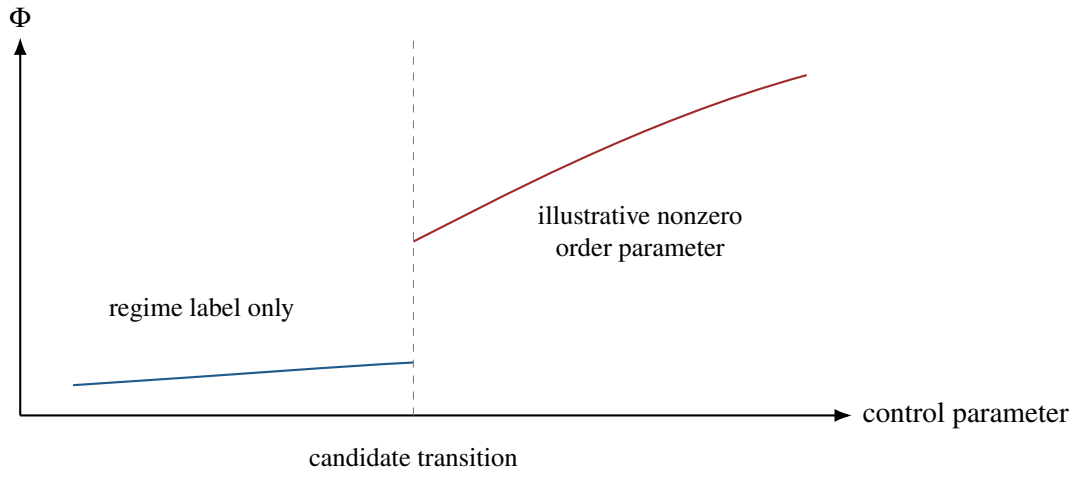


Figure 53.1: Schematic phase criterion. A regime can be smoothly connected to ordinary matter; a phase claim requires additional thermodynamic structure.

Part VI

Axisymmetric General-Relativistic Equilibrium

54

Why Spherical TOV Is Not the Final Equation

The original draft used a spherical anisotropic TOV equation while simultaneously asserting a strongly magnetized axisymmetric geometry. The present formulation treats the spherical TOV equation only as a controlled limiting case. Fully relativistic magnetized stars are naturally described by coupled Einstein-Maxwell equations in stationary axisymmetry, where the electromagnetic field contributes both stress-energy and Lorentz forces. [7, 8]

55

Metric Ansatz and Magnetic Geometry

55.1 Stationary axisymmetry

A convenient quasi-isotropic metric is

$$\begin{aligned} ds^2 = &-N^2\,dt^2 + A^2(dr^2 + r^2\,d\theta^2) \\ &+ C^2 r^2 \sin^2\theta (d\varphi - \omega\,dt)^2, \end{aligned} \tag{55.1}$$

where N, A, C, ω depend on (r, θ). For a nonrotating configuration $\omega = 0$.

55.2 Four-potential

For a nonrotating, purely poloidal field one commonly starts with an azimuthal potential component $A_\varphi(r, \theta)$. A rotating model can require $A_t \neq 0$. Mixed poloidal-toroidal configurations demand a richer current prescription. Fully self-consistent calculations must specify the current function and the exterior matching. [7, 10]

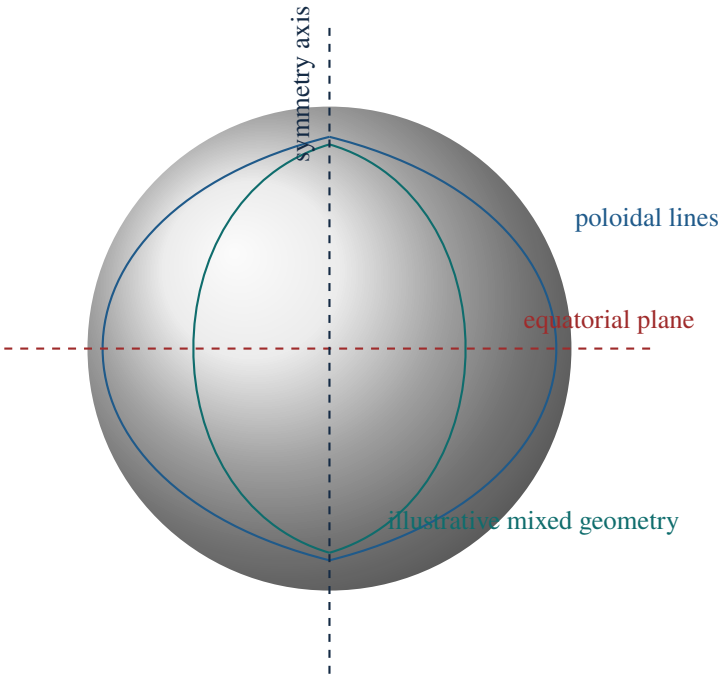


Figure 55.1: Magnetic geometry is part of the physics. The same local field amplitude can lead to different global deformations for different topologies.

56

Einstein-Maxwell-Matter Boundary-Value Problem

The coupled system is

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}, \tag{56.1}$$

$$\nabla_\mu H^{\mu\nu} = J^\nu, \qquad \nabla_{[\alpha} F_{\beta\gamma]} = 0, \tag{56.2}$$

$$\nabla_\mu T^{\mu\nu} = 0, \tag{56.3}$$

plus the explicit microphysical closure, composition constraints, and boundary conditions.

The spatial projection of conservation is

$$h^\alpha{}_\nu \nabla_\mu T^{\mu\nu} = 0, \tag{56.4}$$

which encodes relativistic force balance. Electromagnetic and magnetization forces are already contained in the total tensor, so no extra Lorentz force may be added a second time.

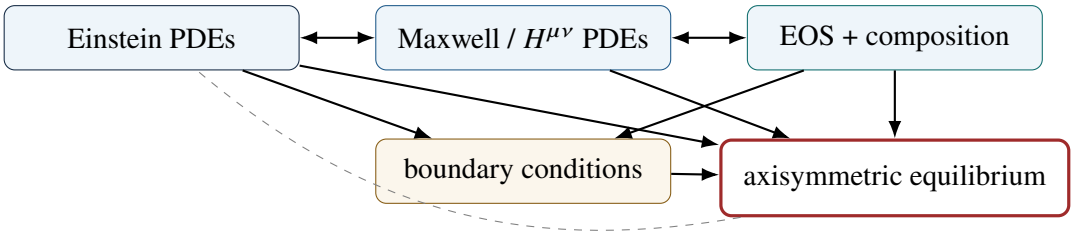


Figure 56.1: Self-consistent equilibrium loop. Geometry, field, microphysics, and boundaries are solved together.

56.1 Boundary conditions

A practical equilibrium calculation must specify regularity at the center, symmetry conditions on the axis, matching conditions across the stellar surface, and asymptotic exterior behavior. The electromagnetic field must be matched to an exterior magnetosphere or electrovacuum model. The current distribution and the field topology are part of the definition of the solution class.

57

Horizon-Free Domain and Compactness

For a spherical object one useful compactness parameter is

$$C = \frac{GM}{Rc^2}, \tag{57.1}$$

with $2C < 1$ as a necessary condition for a surface lying outside the Schwarzschild radius. In a general axisymmetric spacetime, horizon formation and the causal structure are not captured by a single scalar compactness number, so the numerical solution must be checked for the absence of a horizon or trapped-surface condition appropriate to the adopted geometry.

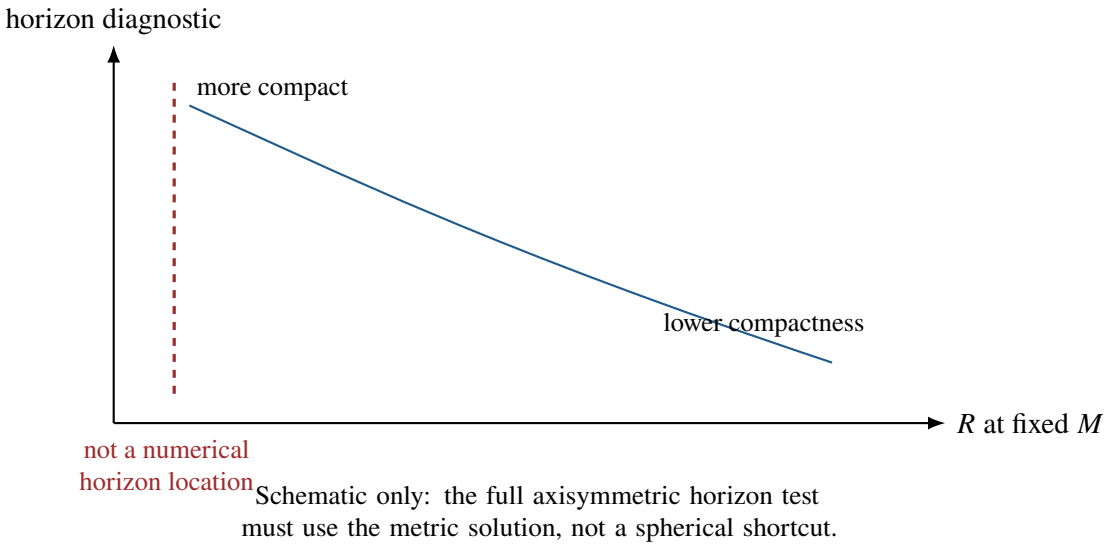


Figure 57.1: Compactness and horizon-free reasoning.

Part VII

Stability, Causality, and Perturbation Theory

58

Local Thermodynamic Consistency

The former claim that three local inequalities constitute “stability iff” is removed. Local conditions can reject obviously pathological thermodynamic states, but they do not prove global dynamical stability.

A selected thermodynamic path should satisfy an appropriate compressibility and susceptibility structure. A schematic causal bound is

$$0 \le c_s^2 = \left(\frac{\partial P}{\partial \epsilon}\right)_{\text{path}} \le 1 \tag{58.1}$$

in units with $c = 1$, but the magnetized problem is anisotropic and the true characteristic speeds must be obtained from the coupled perturbation system.

Negative $P_\parallel$ or $P_\perp$ is a warning sign, not an automatic proof of collapse. The global field geometry and boundary conditions determine whether an equilibrium remains possible.

59

Global Equilibrium Diagnostics

For an equilibrium sequence, monitor at minimum

$$M, \quad M_b, \quad J, \quad E_B, \quad \rho_c, \tag{59.1}$$

and structural quantities such as the equatorial and polar circumferential radii, quadrupole moment, and magnetic ellipticity.

Turning-point arguments can identify candidate stability boundaries along controlled one-parameter sequences, but they do not replace a dynamical mode calculation for strongly magnetized axisymmetric stars. [2]

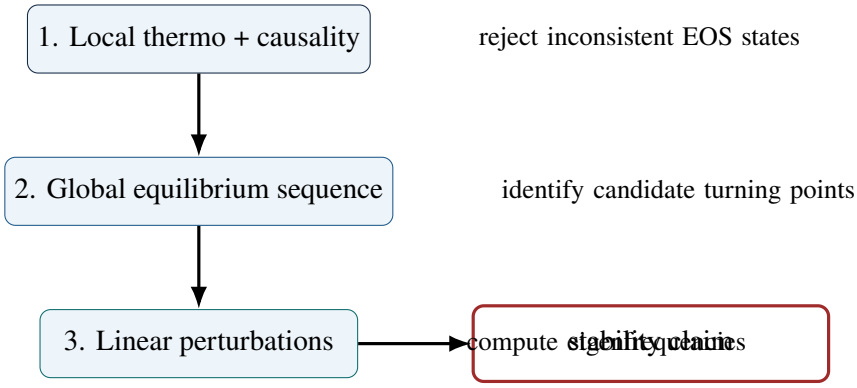


Figure 59.1: Stability hierarchy. Each stage answers a different question.

60

Dynamical Stability

Perturb the full system:

$$\delta G_{\mu\nu} = 8\pi G\, \delta T_{\mu\nu}, \tag{60.1}$$

$$\delta(\nabla_\mu H^{\mu\nu} - J^\nu) = 0, \tag{60.2}$$

$$\delta(\nabla_\mu T^{\mu\nu}) = 0. \tag{60.3}$$

Assume modal time dependence $e^{-i\omega t}$. An instability is signaled by $\Im(\omega) > 0$ under the convention adopted here.

The perturbations contain metric, electromagnetic, and matter degrees of freedom. In rotating systems, the mode set also couples to frame-dragging and non-axisymmetric structure.

61

Mode Classification

Natural mode families include fluid pressure modes, gravity modes, Alfvén-like magnetic modes, rotational inertial modes, and coupled magneto-fluid spacetime modes. The precise spectrum depends on the microphysical closure and the field topology.

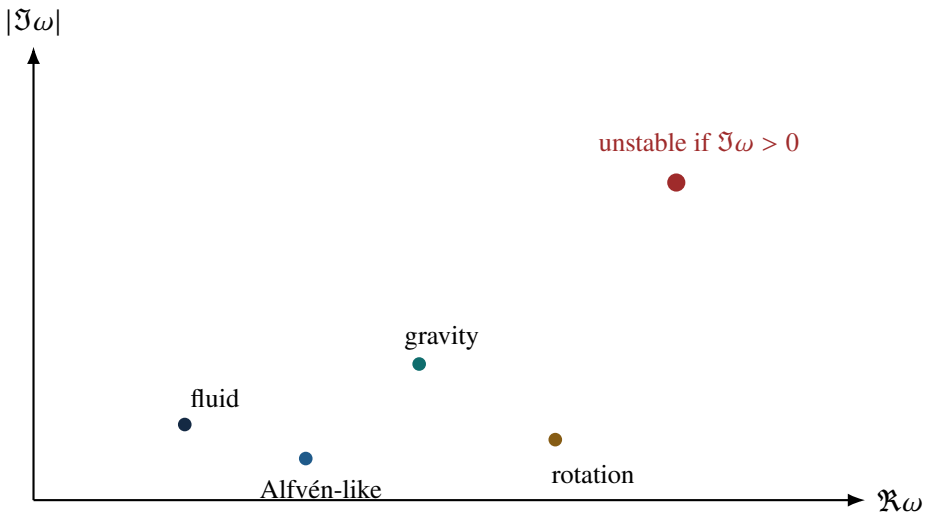


Figure 61.1: Schematic mode spectrum. The red point represents a mode with positive growth rate under the chosen convention.

Part VIII

Observable Consequences

62

Vacuum Birefringence and Polarized Photon Propagation

62.1 Established QED effect

Strong-field QED makes the vacuum behave as a nonlinear optical medium. Two photon polarization eigenmodes can experience different refractive responses in a strong magnetic field. This effect is established in QED and is therefore not, by itself, a unique discovery of HMGQM. [4]

In 2026, observations of the radio magnetar 1E 1547.0−5408 found high X-ray polarization, with phase-averaged polarization degrees reaching about 65% near 2 keV and phase-dependent values approaching 80% in the 2-3 keV band; the reported propagation pattern was consistent with vacuum-birefringence-governed models. [13]

The HMGQM-specific task is more stringent: compute photon propagation through a *self-consistent* curved, magnetized interior and magnetosphere, then quantify the difference from a baseline magnetized neutron-star model with comparable surface observables.

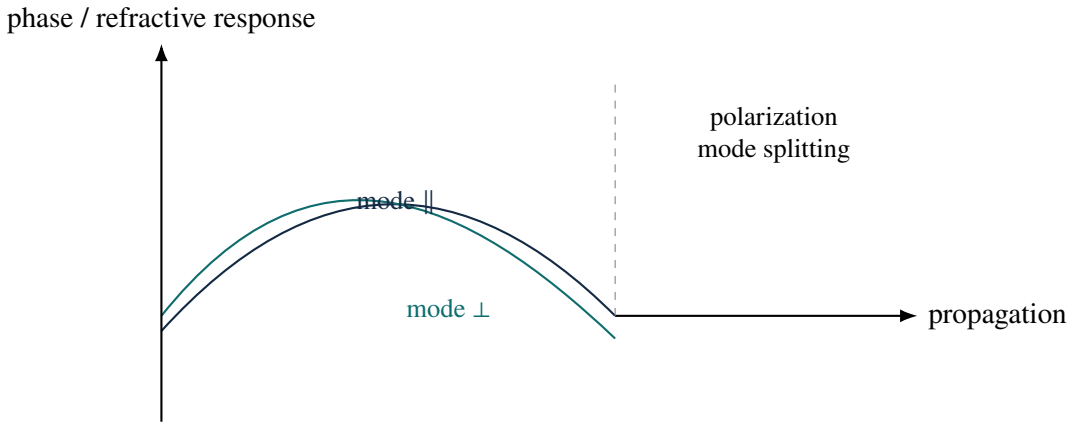


Figure 62.1: Polarization-dependent photon propagation in a magnetized vacuum. The curves are schematic, not numerical predictions.

62.2 Finite-field response

The common weak-field scaling

$$n_{\parallel} - n_{\perp} \propto \alpha \left(\frac{B}{B_{\mathrm{QED}}} \right)^2 \tag{62.1}$$

should be treated as an asymptotic expression, not extrapolated blindly into arbitrary $B \gg B_{\mathrm{QED}}$. Finite-field one-loop treatments retain the full field dependence. Recent work has shown that this distinction can materially alter the predicted accumulated birefringent phase and resonance properties at magnetar-scale fields. [12]

63
Polarized Ray Tracing in Curved Spacetime

The photon problem is a transport problem, not merely a geodesic shift. The pipeline is

$$\begin{aligned} \{g_{\mu\nu}, F_{\mu\nu}\} &\to \text{local QED response} \\ &\to \text{mode-dependent propagation} \\ &\to \text{surface/magnetosphere transfer} \\ &\to \text{detector Stokes parameters.} \end{aligned} \tag{63.1}$$

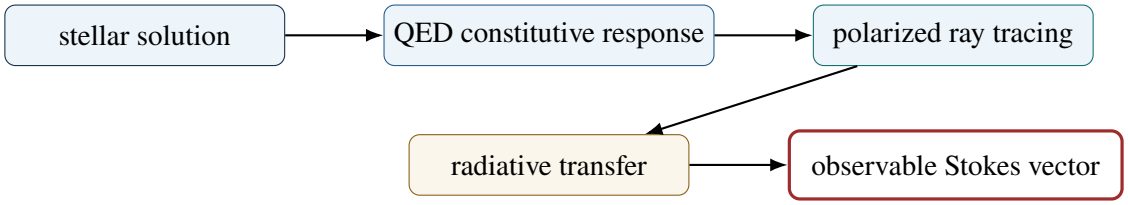


Figure 63.1: Research pipeline for polarization observables.

64

Gravitational-Wave Observables

The previous parameterization $\omega = \omega^{(0)} + \delta\omega(B, \Delta P)$ is retained only as an intuition. A prediction requires solving the perturbed coupled operator

$$\boxed{\mathcal{D}[\delta X; g_{\mu\nu}, F_{\mu\nu}, \Omega] = 0,} \tag{64.1}$$

where δX denotes the perturbations of geometry, electromagnetic field, and matter. The eigenvalues ω_{nlm} become predictions only after boundary conditions and gauge conditions are specified.

64.1 GW170817 and the post-merger caution

GW170817 was a major compact-binary observation, but searches did not detect a post-merger gravitational-wave signal. It therefore should not be plotted as if it supplied a measured post-merger mode frequency. [14]

64.2 Potential model outputs

The relevant quantities include

$$\Delta\omega_{nlm}, \qquad \epsilon_B, \qquad h_0, \qquad \tau_{\mathrm{damp}}, \tag{64.2}$$

plus changes in the mode coupling matrix, polarization, and possible non-axisymmetric sidebands.

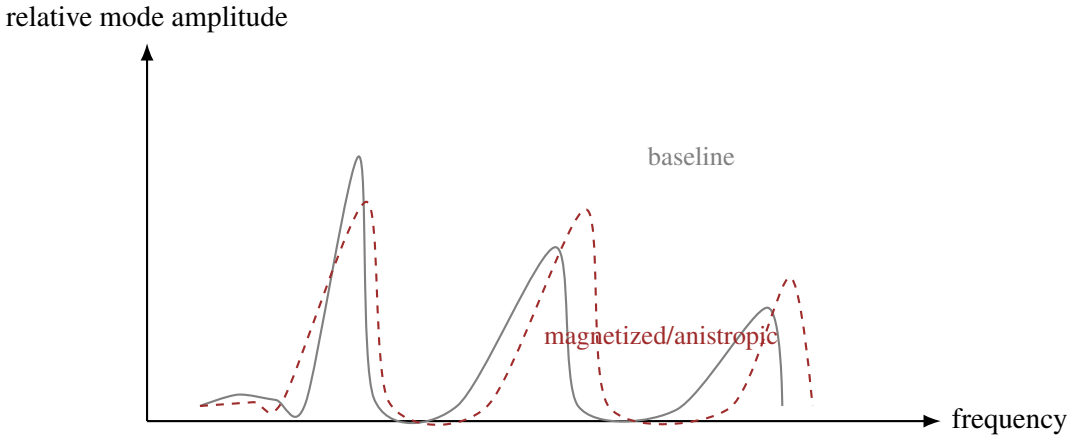


Figure 64.1: Illustrative mode-spectrum shift. Not a fit to LIGO data; the point is that a calculation must predict the shift from the perturbation operator.

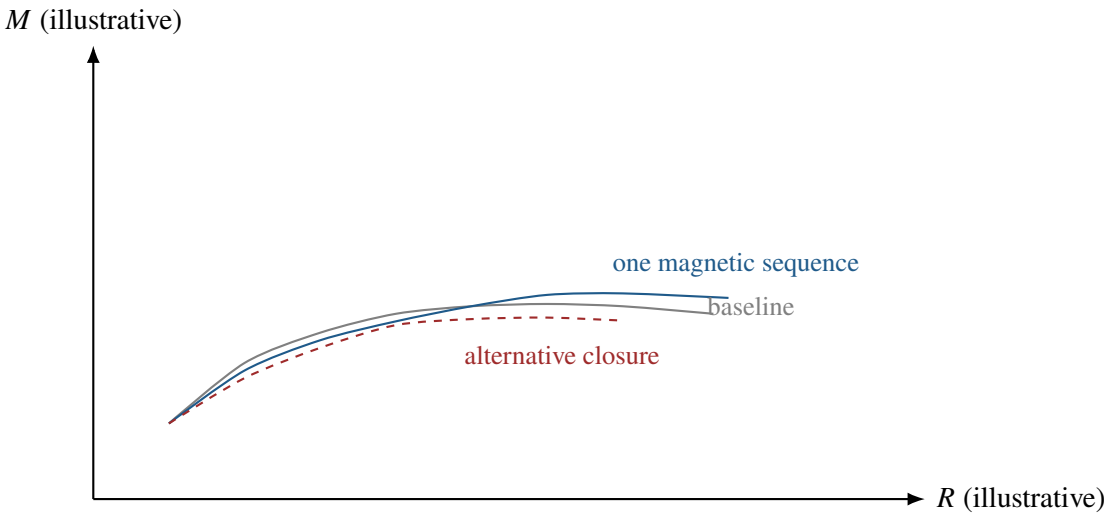
65

Mass-Radius, Tidal, Rotational, and Electromagnetic Diagnostics

For each microphysical EOS and magnetic topology, the framework generates a family of outputs:

$$M(R), \quad \Lambda(M), \quad I(M), \quad E_B(M), \quad \epsilon_B(M), \quad Q(M), \tag{65.1}$$

and polarization/light-curve observables. A difference between the HMGQM calculation and a baseline model is meaningful only after the baseline uses comparable data, field assumptions, and nuisance parameters.



Schematic only. Actual curves must be computed for stated EOS, field topology, composition, and control parameters.

Figure 65.1: Mass-radius comparison as a diagnostic family rather than an assumed HMGQM signature.

Part IX

Numerical Formulation and Research Workflow

66

Dimensionless Formulation

A numerically stable implementation should nondimensionalize the equations. Natural scales can be built from m_e, B_{QED}, the chosen baryon mass scale, and the stellar radius R_*.

For example,

$$\hat{B} = \frac{B}{B_{\mathrm{QED}}}, \qquad \hat{n}_B = \frac{n_B}{n_{\mathrm{sat}}}, \qquad \hat{P} = \frac{P}{P_*}, \qquad \hat{r} = \frac{r}{R_*}. \tag{66.1}$$

This makes convergence studies and parameter-space scans easier to compare across units.

67

Numerical Architecture

A production pipeline can be organized as follows:

1. select a microphysical EOS and particle content;
2. solve charge neutrality and beta equilibrium;
3. evaluate Landau-level sums and AMM corrections;
4. evaluate the finite-field Heisenberg-Euler response;
5. compute Ω, M, ϵ, $P_{\parallel}$, and $P_{\perp}$;
6. construct $T_{\mu\nu}$ and $H^{\mu\nu}$;
7. solve the axisymmetric Einstein-Maxwell boundary-value problem;
8. extract global observables and verify constraints;
9. perturb the equilibrium and solve the eigenvalue problem;
10. ray-trace polarized photons and compare with a baseline model.

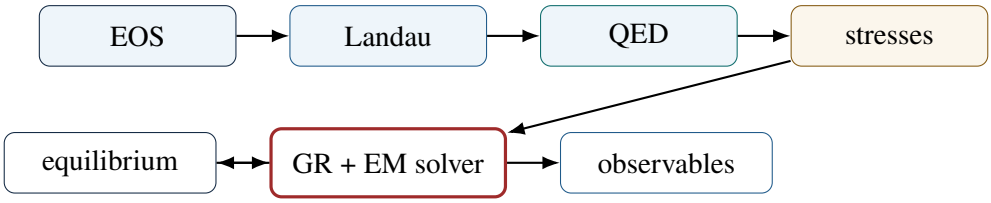


Figure 67.1: Numerical architecture from microphysics to observables.

68

Convergence and Error Control

The numerical solution is not scientifically useful unless key quantities converge under mesh refinement, spectral resolution, Landau-sum truncation tolerance, and nonlinear solver tolerance. Recommended checks include:

- residual norms for Einstein and Maxwell equations;
- violation of $\nabla_\mu T^{\mu\nu} = 0$;
- convergence of global mass and baryon number;
- stability of observables against domain enlargement;
- sensitivity to Landau-level cutoff and interpolation resolution;
- sensitivity to the chosen exterior magnetosphere treatment.

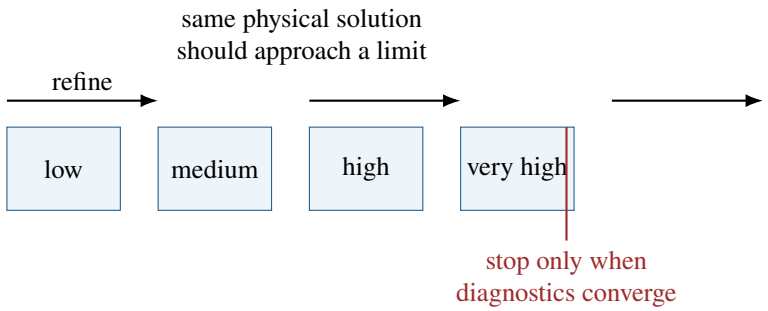
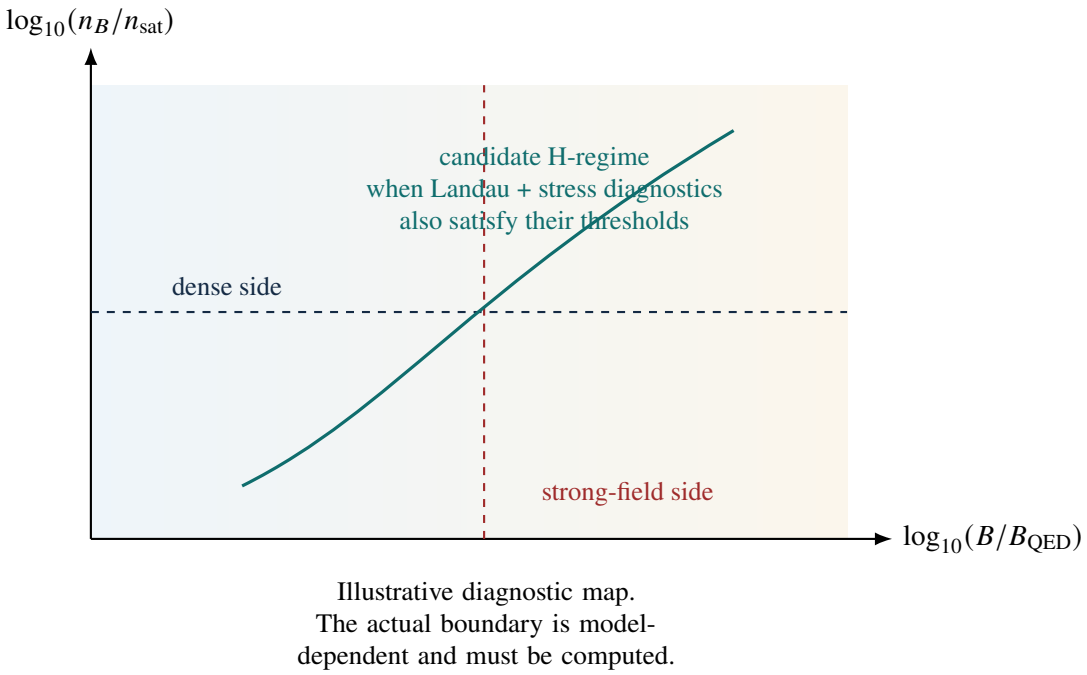


Figure 68.1: Convergence ladder. A stable-looking picture without convergence is not a validated result.

69

Parameter-Space Studies

A useful scan uses at least three axes: field strength χ_B, density n_B/n_{sat}, and Landau-level occupancy. A fourth axis is the field topology.



Illustrative diagnostic map.
The actual boundary is model-dependent and must be computed.

Figure 69.1: H-regime parameter-space map. No fixed universal boundary is assumed.

70

Benchmark Problems

Before attempting full H-Matter calculations, the code should reproduce simpler published limits:

1. non-magnetized TOV for the chosen EOS;
2. weak-field Newtonian magnetohydrostatic equilibrium;
3. axisymmetric relativistic poloidal-field sequences;
4. the zero-temperature Landau-quantized free-fermion gas;
5. the weak-field expansion of the Heisenberg-Euler action;
6. a finite-field QED birefringence benchmark.

Bocquet et al. provide a classical fully relativistic benchmark class for magnetized stars, while modern reviews describe the microphysical Landau and AMM sectors. [7, 10, 11]

71

Reproducibility Protocol

Every numerical paper based on HMGQM should report: code version, EOS name and parameterization, particle content, field topology, central density, central field, rotation rate, numerical resolution, boundary treatment, unit convention, regularization/renormalization convention for QED, and convergence metrics.

A result with no stated microphysical closure is not reproducible. A result with no field topology is not reproducible. A result with no convergence statement is not quantitatively defensible.

72

Microphysics Module Architecture

The microscopic model is intentionally separated into modules so that uncertainties do not become hidden assumptions. A reference implementation may contain a hadronic mean-field sector, a lepton Landau sector, an AMM sector, and the finite-field QED sector.

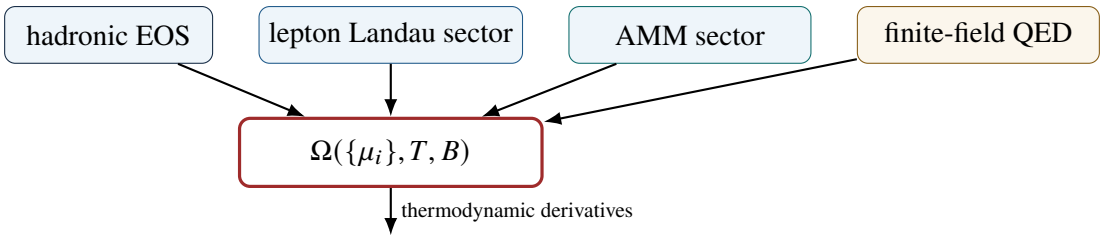


Figure 72.1: Modular microphysics. Each block can be changed or benchmarked without rewriting the global GR structure.

72.1 Reference EOS versus universal HMGQM physics

The framework should distinguish what is universal from what is model-dependent. The covariant coupling structure and stress accounting are universal within the present formulation. The numerical values of effective masses, coupling constants, onset densities, and possible hyperon/quark thresholds are properties of the selected dense-matter model.

Reproducibility rule

An HMGQM result should never be reported only as a function of B. It should specify the EOS, composition, temperature, field geometry, and thermodynamic ensemble. Two microphysical closures can produce different responses at the same field amplitude.

73

Species-Resolved Landau Occupancy

For each charged species, the field response should be summarized by $\nu_{\max,i}$ rather than by a single statement that "matter is quantized." This resolves the species-dependence problem in the original draft.

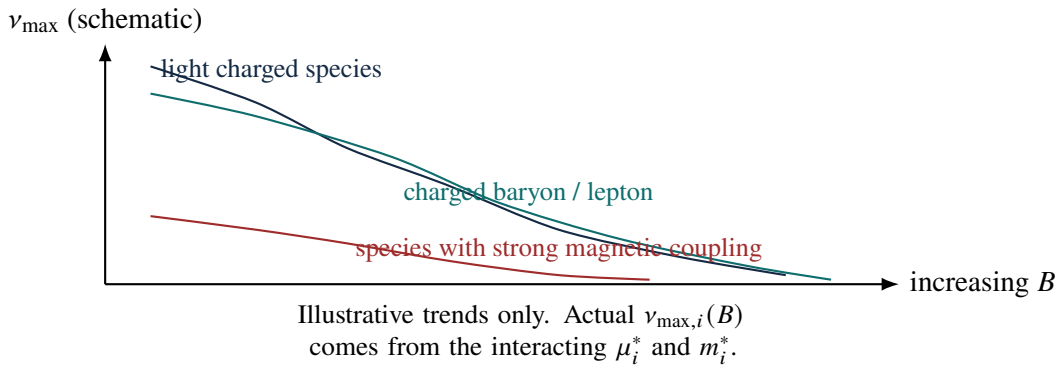


Figure 73.1: Species-resolved Landau occupancy.

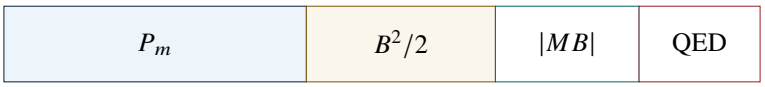
74

Magnetic Stress Budget

The magnetic contribution must be tracked relative to the matter stress. Define a diagnostic stress vector

$$\mathbf{\Pi} = \left(P_m,\ B^2/2,\ |MB|,\ |\Delta\mathcal{L}_{\mathrm{HE}}| \right). \tag{74.1}$$

The purpose is bookkeeping: it makes clear which sector is actually controlling the local balance.



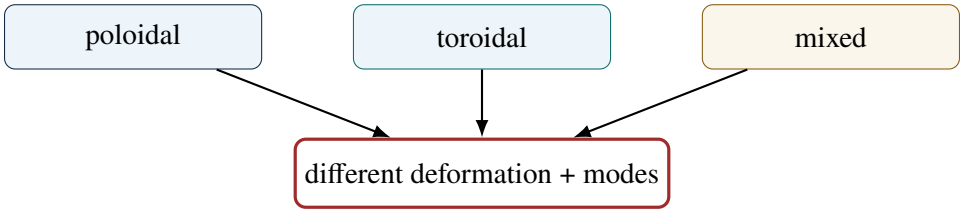
Relative widths are illustrative, not a prediction. In a real solution the widths are computed point-by-point.

Figure 74.1: Local magnetic stress budget as a diagnostic, not a fixed partition.

75

Field-Topology Dependence

Magnetic geometry changes the relation between local field strength and global stress distribution. At minimum, a study should distinguish poloidal, toroidal, and mixed configurations. Rotation adds a second axis whose relative alignment can control time-dependent observables.



A single scalar “central field” cannot encode all of these configurations.

Figure 75.1: Topology family.

76

Surface Matching and Exterior Propagation

The interior star is not an isolated local calculation. At the surface, pressure and field variables must be matched to an exterior electromagnetic solution. For photon observables, the exterior magnetosphere can dominate the propagation distance over which the polarization modes adiabatically separate.

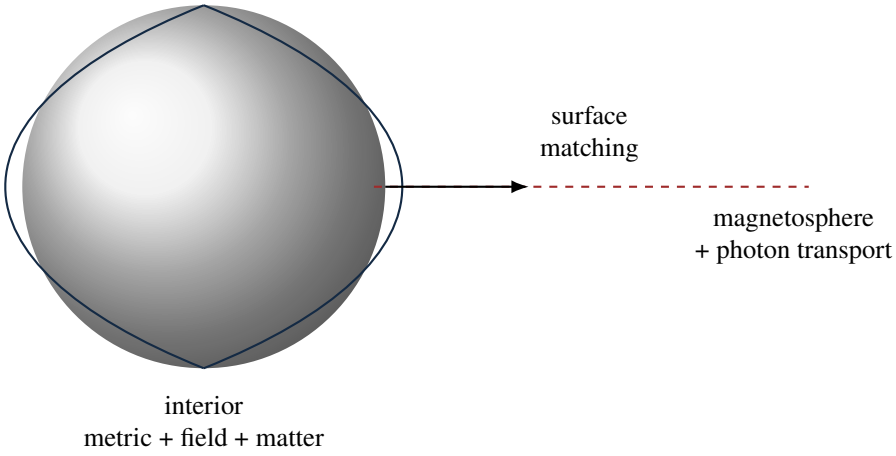
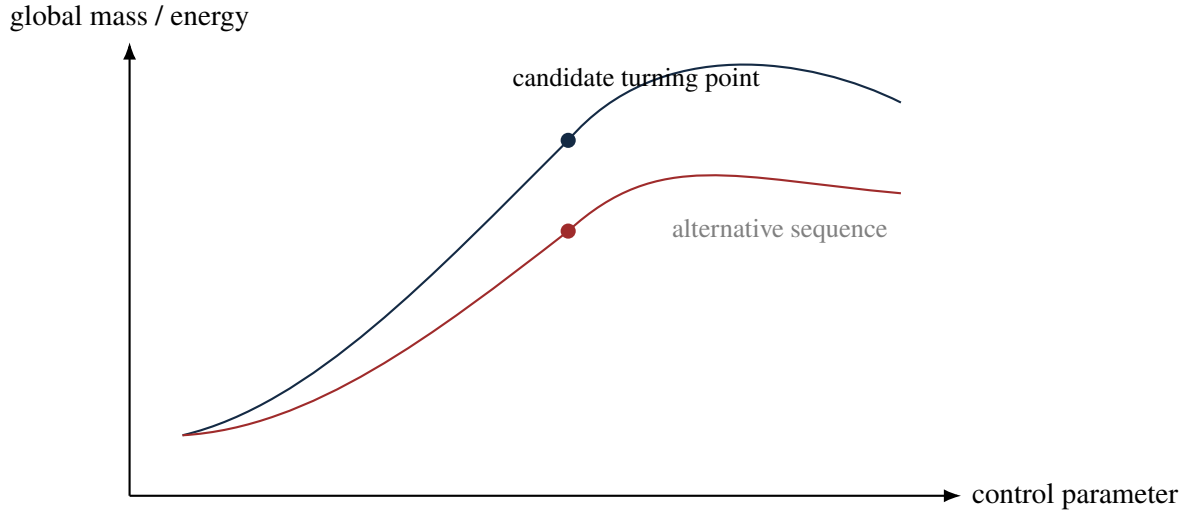


Figure 76.1: Interior-to-exterior matching.

77

Thermodynamic Branch Analysis

A numerical sequence should be parametrized in a controlled way. Useful choices include central baryon density, central chemical potential, baryon mass, magnetic dipole moment, or a prescribed current-function amplitude. One must avoid switching the control parameter mid-sequence.



Turning points can indicate a candidate stability boundary, but they do not prove dynamical stability.

Figure 77.1: Schematic equilibrium-branch analysis.

78

Uncertainty Budget

A complete HMGQM result has multiple layers of uncertainty: dense-matter EOS, field topology, QED truncation, numerical resolution, exterior transport, and observational nuisance parameters. A robust claim must propagate these uncertainties rather than selecting the combination that maximizes a desired signal.

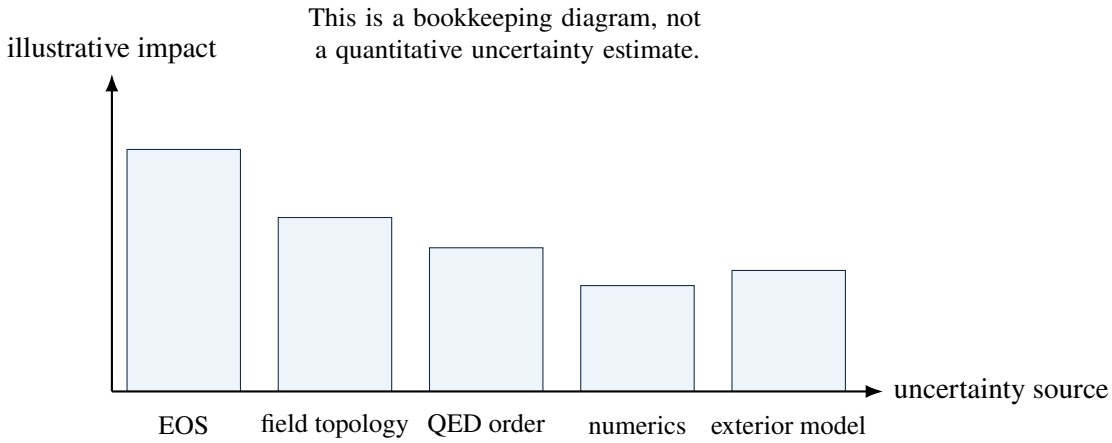


Figure 78.1: Suggested uncertainty budget.

79

Verification Pyramid

Each layer should be verified before the next is trusted.

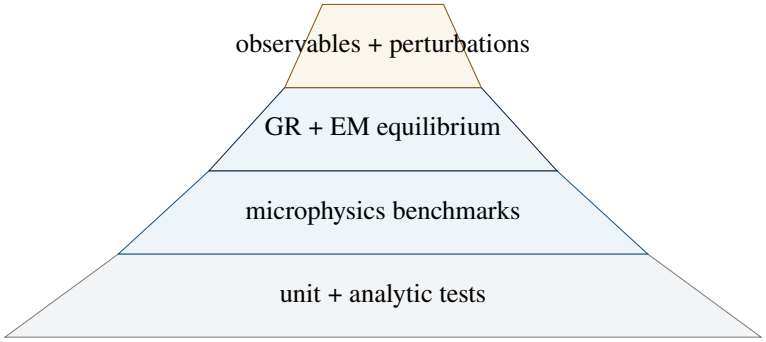


Figure 79.1: Verification pyramid for a reproducible implementation.

80

Multi-Messenger Consistency Triangle

The strongest future test of HMGQM would connect interior structure to more than one messenger using the same physical solution.

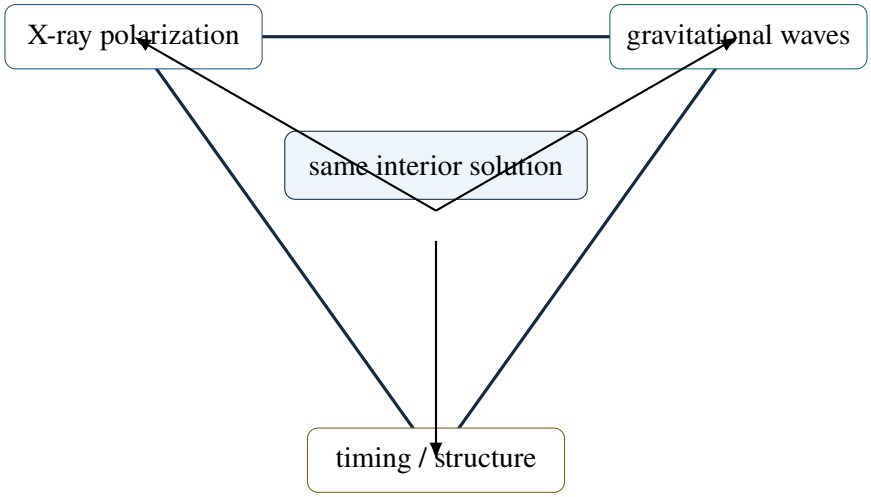


Figure 80.1: Multi-messenger consistency triangle. Correlated predictions are stronger than isolated phenomenology.

81

Model-Selection Logic

A statistically preferred fit is not the same as a physically established phase. The model-selection chain should therefore be explicit:

$$\begin{aligned} &\text{physical consistency} \rightarrow \text{numerical convergence} \\ n \rightarrow\ &\text{predictive improvement} \rightarrow \text{parameter robustness} \\ n \rightarrow\ &\text{independent evidence.} \end{aligned} \tag{81.1}$$

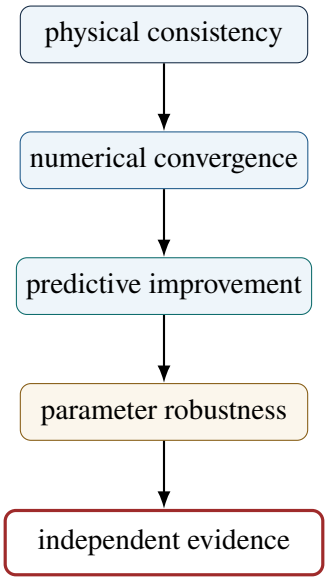


Figure 81.1: Model-selection logic. Each step narrows the space of defensible claims.

82

A Reproducibility Manifest

A release package for HMGQM calculations should contain:

1. machine-readable EOS tables or the exact EOS parameterization;
2. unit and convention definitions;
3. field topology and current prescription;
4. numerical resolution and solver tolerances;
5. QED integration and derivative tolerances;
6. equilibrium sequences and raw global quantities;
7. perturbation eigenvalues and mode eigenfunctions;
8. ray-tracing assumptions and exterior model;
9. scripts sufficient to regenerate every published figure;
10. a machine-readable manifest identifying software versions and dependencies.

83

Worked Symbolic Consistency Check

At fixed composition, the Gibbs-Duhem relation provides a useful local diagnostic. For a homogeneous system at $T = 0$,

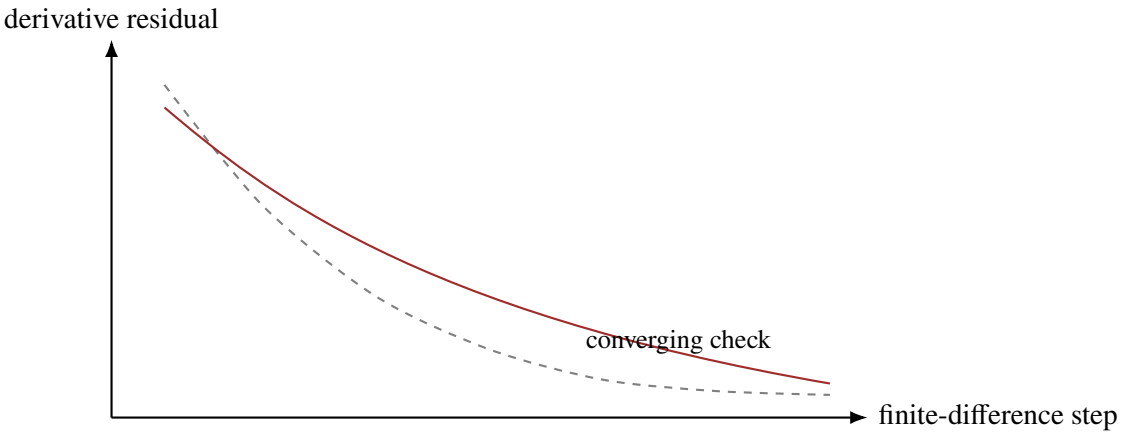
$$dP_m = \sum_i n_i \; d\mu_i + M \; dB, \tag{83.1}$$

under the adopted thermodynamic convention. The numerical derivative of the code-generated pressure should agree with the direct derivative of the grand potential. A mismatch indicates an inconsistency in the EOS, magnetization, or interpolation layer.

The stress anisotropy satisfies

$$\Delta P = B^2 - MB. \tag{83.2}$$

Hence any state reported as H-Matter must show the actual magnitude of this quantity and not merely state that anisotropy is present.



Illustrative numerical verification curve. In a valid code the residual should decrease toward the discretization floor.

Figure 83.1: Symbolic consistency-check concept for thermodynamic derivatives.

Part X

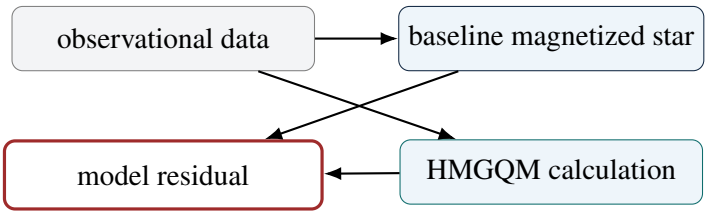
Falsifiability and Scientific Tests

84

Observational Discipline

84.1 Baseline comparison principle

The theory should be tested against a baseline that uses the same external observables and as many shared nuisance parameters as possible. A distinctive HMGQM claim exists only in the residual difference.



The scientifically interesting quantity is often the difference between two self-consistent models, not the absolute observable alone.

Figure 84.1: Baseline-comparison principle for falsifiable predictions.

85

Failure Conditions

The framework should be considered unsuccessful as a physical model if one or more of the following occurs:

1. no numerically converged equilibrium exists in the claimed regime for any plausible microphysical closure;
2. the claimed strong-field effects remain perturbatively tiny everywhere in the self-consistent solution;
3. the model violates thermodynamic consistency, Maxwell identities, or energy-momentum conservation;
4. all apparent observational shifts disappear within uncertainty and baseline-model freedom;
5. no phase criterion is found, if a stronger “new phase” claim is retained;
6. the perturbation spectrum shows generic instability throughout the supposed H-Matter region.

86

Potentially Positive Evidence

Evidence that would strengthen the framework includes a robustly converged new equilibrium branch, a non-analytic thermodynamic feature correlated with the H-regime, a stable range of configurations with significant anisotropic stress, and a multi-messenger observable difference that remains after nuisance-parameter marginalization.

A particularly strong outcome would be a correlated signature: for example, an EOS-induced change in equilibrium structure accompanied by a predictable shift in polarized X-ray transport and a distinct gravitational-wave mode response, all computed from the same interior field configuration.

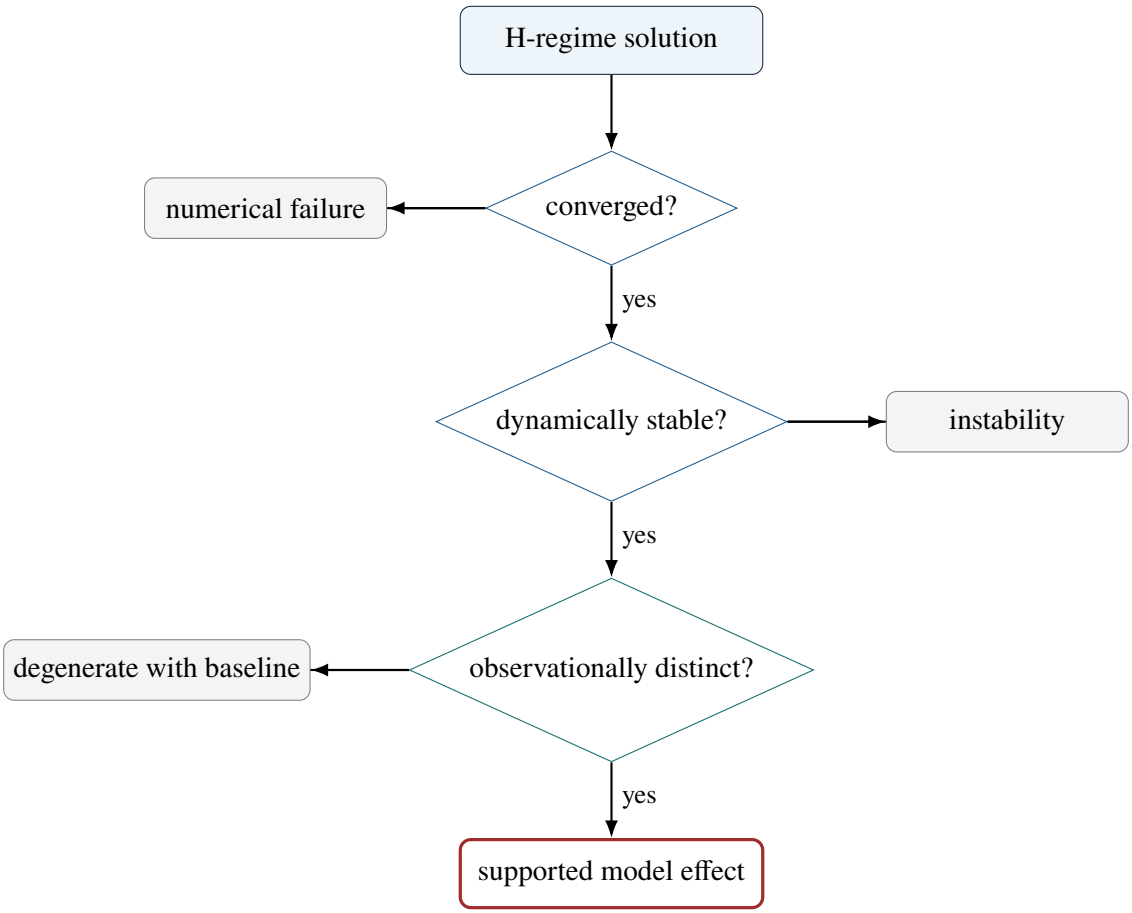


Figure 86.1: Decision tree for turning an H-regime calculation into defensible scientific evidence.

87

Research Program

The clean research program is:

1. select a baseline microphysical EOS;
2. implement exact Landau quantization and AMM terms;
3. implement finite-field one-loop QED response;
4. validate local thermodynamics;
5. solve axisymmetric Einstein-Maxwell equilibrium;
6. build controlled parameter sequences;
7. solve perturbations and mode spectra;
8. compute polarized photon transport;
9. compare against baseline magnetized-star models;
10. only then decide whether the H-Regime warrants a new-phase interpretation.



Each stage has its own verification suite. No later stage is considered validated until the earlier stage converges.

Figure 87.1: End-to-end research pipeline.

Part XI

Conclusions and Final Scientific Position

88

Integrated Theory Statement

The complete HMGQM framework can be summarized compactly by the action

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \frac{1}{4}F^2 + \Delta\mathcal{L}_{\rm HE} + \mathcal{L}_{\rm matter}\right], \tag{88.1}$$

with

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}, \qquad \nabla_\mu H^{\mu\nu} = J^\nu, \qquad \nabla_{[\alpha}F_{\beta\gamma]} = 0, \qquad \nabla_\mu T^{\mu\nu} = 0, \tag{88.2}$$

plus chemical equilibrium, charge neutrality, a specified dense-matter closure, and boundary conditions.

The local EOS is generated from $\Omega(\{\mu_i\}, T, B)$, with

$$M = -\partial_B \Omega, \tag{88.3}$$

$$\epsilon = \Omega + \sum_i \mu_i n_i + B^2/2, \tag{88.4}$$

$$P_\parallel = -\Omega - B^2/2, \qquad P_\perp = -\Omega - MB + B^2/2, \tag{88.5}$$

and charged-particle densities given by the Landau sum.

89

Limiting Cases

89.1 Zero-field limit

As $B \to 0$, Landau sums approach the continuum momentum integral, $M \to 0$, and $P_{\parallel} \to P_{\perp}$ under the chosen convention. The star approaches an isotropic relativistic configuration described by the chosen EOS.

89.2 Weak-field QED limit

For $B \ll B_{\mathrm{QED}}$, the finite-field Heisenberg-Euler response reduces to its familiar low-field expansion. This provides a check, not the defining H-regime formula. [4]

89.3 Many-Landau-level limit

As $\nu_{\max} \to \infty$, the Landau sum approaches the continuum density of states and the quantization correction becomes small relative to the unmagnetized matter result.

89.4 Weak magnetic-stress limit

When $B^2/2 \ll P_m$, global magnetic deformation becomes perturbative, and the axisymmetric solution should approach the corresponding weak-field sequence.

89.5 Newtonian limit

At weak compactness and low velocity, the Einstein-Maxwell system must reduce to magnetohydrostatic balance with Maxwell stress.

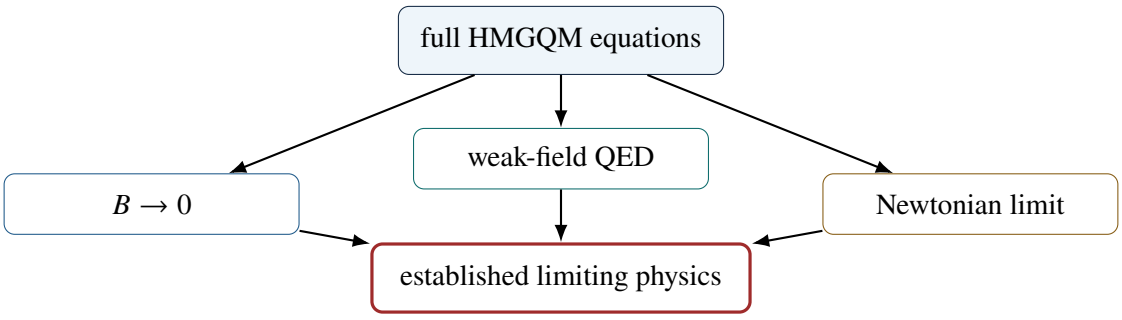


Figure 89.1: Required limiting-case consistency.

90

Conclusion

The original HMGQM proposal had a useful physical intuition but mixed several mathematical regimes that could not safely be claimed at once. The current consolidated formulation keeps the central idea and removes those inconsistencies. HMGQM Theory v3.1 is therefore best understood as a research framework for an extreme magneto-relativistic regime, not as a finished replacement for established physics.

The core mathematical architecture is now explicit: a covariant action, finite-field Heisenberg-Euler QED, a microphysically specified dense-matter grand potential, Landau quantization, magnetization, a single total stress-energy tensor, axisymmetric Einstein-Maxwell equilibrium, and a full perturbation program. This structure is compatible with the known direction of research on strongly magnetized compact stars, where magnetic deformation and microphysical quantization require self-consistent treatment. [7, 8, 10, 11]

The most interesting unresolved question is no longer whether a strong magnetic field can make a pressure anisotropy. That is already expected. The unresolved question is whether the fully coupled system admits a qualitatively new stable branch or thermodynamic phase, and whether that branch produces observable differences that survive comparison with conventional magnetized-star models.

That question is scientifically meaningful precisely because it is falsifiable. A failed convergence test, generic dynamical instability, or observational degeneracy would constrain or reject the stronger HMGQM claims. A robust thermodynamic transition, stable new equilibrium branch, and multi-messenger signature would justify a stronger interpretation.

Final scientific position

HMGQM Theory v3.1 proposes that an extreme, strongly magnetized dense-matter regime can require the simultaneous treatment of Landau-quantized microphysics, finite-field QED vacuum response, electromagnetic stress, anisotropic principal pressures, and axisymmetric general-relativistic equilibrium. The label H-Matter is a regime hypothesis. It becomes a claim of a genuinely new phase only after a non-continuous thermodynamic or order-parameter signature is derived.

91

What This Theory Does Not Claim

- It does not claim a complete quantum theory of gravity.
- It does not introduce a new fundamental particle or interaction.
- It does not claim that all baryonic matter becomes one-dimensional at $B \gg B_{\mathrm{QED}}$.
- It does not claim that vacuum birefringence is unique to HMGQM.
- It does not claim that anisotropy alone proves a new thermodynamic phase.
- It does not claim that GW170817 or GW190814 already confirms HMGQM.
- It does not claim laboratory synthesis or engineering applications.
- It does not claim experimental verification at the present stage.

Closing Statement

The purpose of this monograph is not to declare that a new state of nature has already been discovered. Its purpose is to make the HMGQM hypothesis precise enough that other researchers can test it, reproduce its calculations, criticize its assumptions, and improve or falsify it. That is the point at which a personal theoretical idea can begin to become a scientific research program.

Appendices

A

Landau-Level Derivation Details

For a charged relativistic particle in a uniform magnetic field, choose a local frame with $\boldsymbol{B} = B\hat{z}$. The transverse kinetic momenta obey the Landau algebra, leading to quantized transverse energy $2\nu|qB|$. The remaining longitudinal momentum p_z stays continuous. Summing over the allowed levels gives

$$\sum_{\nu} g_{\nu} \int \frac{dp_z}{2\pi} \frac{|qB|}{2\pi}. \tag{A.1}$$

At fixed chemical potential only states with $E_{\nu}(p_z) \leq \mu^*$ are occupied. This directly produces the cutoff $\nu_{\max}$ and the density formula used in the main text.

A.1 Lowest-Landau-level criterion

The condition $\nu_{\max} = 0$ is equivalent to

$$(\mu^*)^2 - (m^*)^2 < 2|q|B. \tag{A.2}$$

This criterion is species-specific and should always be evaluated after the interacting chemical potential and effective mass have been determined.

B

Finite-Field QED Checks

The finite-field proper-time integral should satisfy the following numerical checks:

1. renormalized vacuum normalization at $B = 0$;
2. smooth approach to the low-field series;
3. stable evaluation of the large-B asymptotic regime;
4. consistent numerical derivatives when computing magnetization;
5. bounded sensitivity to quadrature cutoff and mesh refinement.

The low-field consistency expansion takes the form

$$\Delta\mathcal{L}_{\mathrm{HE}} \sim \frac{\alpha^2}{90m_e^4}\left[(F_{\mu\nu}F^{\mu\nu})^2 + \frac{7}{4}(F_{\mu\nu}\tilde{F}^{\mu\nu})^2\right], \qquad B \ll B_{\mathrm{QED}}, \tag{B.1}$$

up to convention-dependent normalization. The document never uses this truncation as the defining strong-field response. [4]

C

Stress Tensor Identities

Starting from

$$T_{\mu\nu} = \epsilon u_\mu u_\nu + P_\perp h_{\mu\nu} + (P_\parallel - P_\perp) b_\mu b_\nu, \tag{C.1}$$

we obtain in the comoving orthonormal basis

$$T_{\hat{0}\hat{0}} = \epsilon, \quad T_{\hat{z}\hat{z}} = P_\parallel, \quad T_{\hat{x}\hat{x}} = T_{\hat{y}\hat{y}} = P_\perp. \tag{C.2}$$

The isotropic limit is the submanifold $P_\parallel = P_\perp$.

D

Dimensionless Research Variables

A compact computational state vector may be represented as

$$\boldsymbol{y} = \left(\hat{n}_B, \hat{B}, \hat{\mu}_n, \hat{\mu}_p, \hat{\mu}_e, \hat{\mu}_\mu, \hat{M}, \hat{P}_\parallel, \hat{P}_\perp\right). \tag{D.1}$$

Derived diagnostic ratios include

$$\mathcal{A}_P = \frac{|P_\perp - P_\parallel|}{\max(|P_\perp|, |P_\parallel|, P_0)}, \tag{D.2}$$

where P_0 is a numerical floor used only to avoid division by a vanishing denominator.

E

Suggested Numerical Pseudocode

```
for equilibrium_parameter in sequence:
   initialize_density_and_field()
   for nonlinear_iteration in range(max_iter):
      solve_beta_equilibrium_and_charge_neutrality()
      build_landau_sums_and_AMM_terms()
      evaluate_HE_finite_field_response()
      compute_Omega_M_energy_pressures()
      construct_Tmunu_and_Hmunu()
      solve_axisymmetric_Einstein_Maxwell_step()
      enforce_boundary_conditions()
      check_conservation_and_residuals()
      if converged():
         break
   extract_global_observables()
   if converged():
      solve_linear_perturbations()
      compute_polarized_photon_transport()
      compare_against_baseline()
```

F

Benchmark Table

Benchmark	Expected behavior	Purpose
$B \to 0$	isotropic EOS + TOV-like limit	verify thermodynamic recovery
$\nu_{\max} \to \infty$	continuum momentum integral	verify Landau summation
$B \ll B_{\mathrm{QED}}$	low-field QED series	verify renormalized QED implementation
weak compactness	Newtonian magnetohydrostatics	verify gravity reduction
axisymmetric poloidal field	published relativistic magnetic-star behavior	verify global solver
conservation residual	tends to discretization floor	verify self-consistency
mode test	stable/unstable sign matches benchmark	verify perturbation solver

G

Observational Context Notes

G.1 Magnetar polarization

The 2026 Nature study of 1E 1547.0−5408 reported high polarization and argued that vacuum-birefringence-governed magnetospheric propagation can explain the observed phase and energy dependence. [13] This motivates a stronger HMGQM test: the model should predict deviations relative to a baseline that already includes QED propagation.

G.2 GW170817

The LIGO Scientific Collaboration reports that searches did not detect a post-merger gravitational-wave signal for GW170817. [14] Therefore a HMGQM figure should not place GW170817 as a direct measured post-merger frequency point.

H

Reference Notes and Research Basis

This framework deliberately anchors itself to prior literature. The cited sources include pedagogical and technical treatments of the Heisenberg-Euler action, strong-field asymptotics, magnetized dense matter, relativistic magnetic-star equilibrium, and current magnetar polarization observations. The 2026 references are particularly relevant because they update the observational and microphysical context at the time of this edition.

I

Expanded Comparison with the Original 23-Page Draft

The original draft is retained as the conceptual ancestor. Its strongest ideas were the identification of anisotropic pressure, vacuum participation, magnetic geometry, and astrophysical observables. Its main weaknesses were regime mismatch and under-specified mathematics. [1]

Element	Original form	v3.1 form
Gravity	strong curvature + speculative quantum geometry	classical GR with explicit curvature diagnostics
QED	finite polynomial plus strong-field claim	finite-field one-loop Heisenberg-Euler action
Matter	generic dense fermion fluid	explicit EOS module + species content
Landau structure	qualitative / incomplete	exact level sum for charged species
Stress tensor	anisotropic ansatz + separate vacuum term	one variational $T_{\mu\nu}$
Equilibrium	modified spherical TOV	stationary axisymmetric Einstein-Maxwell-matter PDEs
Stability	three local inequalities	local + global + dynamical hierarchy
New phase	inferred from anisotropy	requires thermodynamic transition/order parameter
Observables	proposed polarization/GW signatures	baseline-subtracted, model-derived observables
Evidence	conceptual plots	converged predictions with uncertainty control

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References

Bibliography

[1] A. Kushwaha, *HDMGQM Theory: H-Matter - Hyper-Dense Magneto-Gravitational Quantum Matter Theory*, personal exploratory draft, 23 pages (2026). See the uploaded original draft and its stated scope, assumptions, equations, and author note. Author portfolio.

[2] A. Kushwaha, *Revised HMGQM Theory: A Self-Consistent Revision*, Version 2.0, October 2026. Internal revision used as the immediate mathematical basis of this edition.

[3] A. Kushwaha, author description in the original HMGQM draft, pp. 21-23 (2026).

[4] G. V. Dunne, "Heisenberg-Euler Effective Lagrangians: Basics and Extensions," arXiv:hep-th/0406216 (2004). arXiv link.

[5] F. Karbstein, "All-Loop Result for the Strong Magnetic Field Limit of the Heisenberg-Euler Effective Lagrangian," Phys. Rev. Lett. 122, 211602 (2019), with 2024 erratum. APS link.

[6] E. J. Ferrer and A. Hackebill, "Equation of State of a Magnetized Dense Neutron System," Universe 5, 104 (2019). Journal link.

[7] M. Bocquet, S. Bonazzola, E. Gourgoulhon, and J. Novak, "Rotating neutron star models with magnetic field," Astron. Astrophys. 301, 757 (1995). arXiv link.

[8] C. Y. Cardall, M. Prakash, and J. M. Lattimer, "Effects of strong magnetic fields on neutron star structure," Astrophys. J. 554, 322-339 (2001). arXiv link.

[9] J. A. Pons and D. Viganò, review material on neutron-star crust physics and strong magnetic fields. Living Reviews in Relativity.

[10] M. Sinha and V. B. Thapa, "Dense Matter and Compact Stars in Strong Magnetic Fields," arXiv:2604.22680 (2026). arXiv link.

[11] W. German, T. Mello, J. P. W. Diener, and G. C. Hillhouse, "Effects of Strong Magnetic Fields on the Equation of State and Mass-Radius Structure of Hyperonic Neutron-Star Matter with Anomalous Magnetic Moments," arXiv:2609.02441 (2026). arXiv link.

[12] S. Abbassi, F. A. Chishtie, and S. R. Valluri, "Finite-Field QED Corrections to Vacuum Birefringence and Magnetar Polarization Transport," arXiv:2607.06422 (2026). arXiv link.

[13] R. E. Stewart et al., "Vacuum birefringence and the polarized X-ray emission from a radio magnetar," Nature 656, 590-594 (2026). Nature link.

[14] LIGO Scientific Collaboration, "Searching for the neutron star or black hole resulting from GW170817," science summary, and related long-duration search. LIGO link.

[15] J. S. Heyl and N. J. Shaviv, "QED and the High Polarization of the Thermal Radiation from Neutron Stars," arXiv:astro-ph/0203058 (2002). arXiv link.

[16] A. Broderick, M. Prakash, and J. M. Lattimer, "The Equation of State of Neutron-Star Matter in Strong Magnetic Fields," arXiv:astro-ph/0001537 (2000). arXiv link.

[17] A. Mastrano, P. D. Lasky, and A. Melatos, "Neutron star deformation due to multipolar magnetic fields," arXiv:1306.4503 (2013). arXiv link.

[18] LIGO Scientific Collaboration, “Another long hard look for the remnant of GW170817,” science summary. LIGO link.

[19] R. Taverna et al., “The long quest for vacuum birefringence in magnetars: 1E 1547.0-5408 and the elusive smoking gun,” Astrophysical Journal Letters (2026). Journal link.

[20] G. Kasza, J. Takatsy, and G. Wolf, “Astrophysical constraints on the cold equation of state of the strongly interacting matter,” European Physical Journal Special Topics (2026). Journal link.